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Consciousness, Memory, and Anesthesia

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KEY POINTS

- Mechanisms of consciousness and memory, and their interruption by general anesthetics, are important scientific problems that have clinical relevance for the practice of anesthesiology.
- Consciousness is characterized by both wakefulness (i.e., the brain being aroused) and awareness (i.e., subjective experience).
- Anesthetics act at structures in the brainstem, hypothalamus, and basal forebrain that regulate sleep-wake states, which may account for loss of wakefulness.
- Anesthetics disrupt connectivity and communication across cortical and thalamocortical networks, which may account for loss of awareness.
- Memory can be subdivided into explicit (conscious) and implicit (unconscious) recall; an example of explicit episodic recall is remembering a surgical event.
- Suppression of explicit episodic recall is one of the most potent effects of most general anesthetics.
- Effects on the hippocampus, amygdala, and prefrontal cortex—as well as the connectivity of these structures—may account for anesthetic-induced amnesia, even before loss of consciousness.

Introduction

SCIENTIFIC AND CLINICAL IMPORTANCE

Consciousness and memory are among the most fascinating and complex subjects in all of science. The richness of human consciousness and memory—and the ability to express this richness in language—is a defining characteristic of *homo sapiens*. Consciousness and memory also have clinical relevance for the anesthesiologist; together, the experience and explicit episodic recall of surgical events is known as the problem of “intraoperative awareness.” When formally assessed, this complication occurs in approximately 1 to 2 cases per 1000¹⁻³ and is associated with a high incidence of posttraumatic stress disorder (PTSD).^{4,5} The incidence of intraoperative consciousness without recall is substantially higher.⁶ To advance the field of perioperative brain monitoring, a detailed understanding of the neurobiology of consciousness, memory, and anesthesia is required.

Consciousness

DEFINITIONS

The field of consciousness studies has been plagued by the indiscriminate use of the term “consciousness.” When we refer to consciousness, we mean *subjective experience*. In simple terms, it is what we lose when we have dreamless sleep and what we regain again in the morning upon awakening. There are, however, several important definitions and distinctions that should be considered.

1. **Awareness:** Cognitive neuroscientists and philosophers use the term “awareness” to mean *only* subjective experience. In clinical anesthesiology, we (inaccurately) use the term “awareness” to include *both* consciousness *and* explicit episodic memory⁷ (the taxonomy of memory will be discussed in the next major section of the chapter).
2. **Connected versus disconnected consciousness:** Connected consciousness is the experience of environmental stimuli (such as surgery), whereas disconnected consciousness is an endogenous experience (such as a dream state).⁸
3. **Consciousness versus responsiveness:** An individual may fully experience a stimulus (such as the command “Open your eyes!”) but not be able to respond (as when a patient is paralyzed but conscious during surgery).⁹

There have been a number of theories proposed to explain the mechanisms of consciousness and general anesthesia. Advances in neuroscience, however, have enabled us to move beyond speculative frameworks and focus on a systems-based approach to both subjects.¹⁰ The remainder of this section on consciousness adopts such an approach by discussing (1) brainstem and hypothalamic nuclei regulating the sleep-wake cycle (and therefore arousal states) (Figs. 9.1 and 9.2); (2) the role of the thalamus in consciousness and anesthesia; (3) cortical-subcortical connectivity, with a focus on the thalamocortical system, which is thought to mediate the experiential component of consciousness; and (4) corticocortical communication.

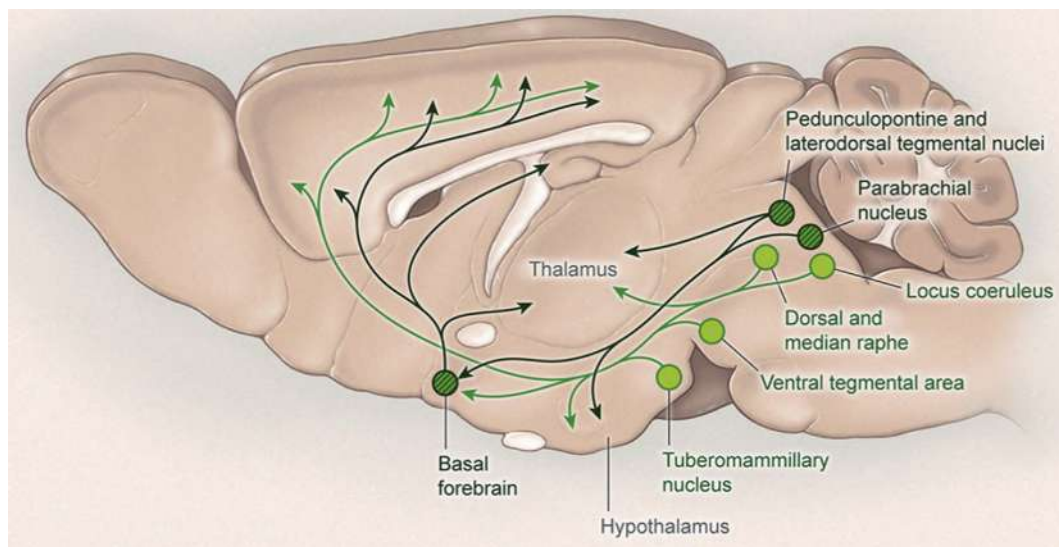


Fig. 9.1 Neurobiology of wakefulness. Multiple neurochemical systems in subcortical regions (shown here in rodent brain) promote arousal and activation of the cortex. Monoaminergic neurons (light green) in the rostral brainstem and caudal hypothalamus innervate the cortex as well as many subcortical regions including the hypothalamus and thalamus. These monoaminergic regions include noradrenergic neurons (locus coeruleus), serotonergic neurons (dorsal and median raphe nuclei), dopaminergic neurons (ventral tegmental area), and histaminergic neurons (tuberomammillary nucleus). Wake-promoting signals also arise from cholinergic regions (dark green with hatching), including the pedunclopontine and laterodorsal tegmental nuclei and basal forebrain. General anesthetics have been demonstrated to suppress many of these regions. (Redrawn from Scammell TE, Arrigoni E, Lipton JO. Neural circuitry of wakefulness and sleep. *Neuron*. 2017;93[4]:747–765.)

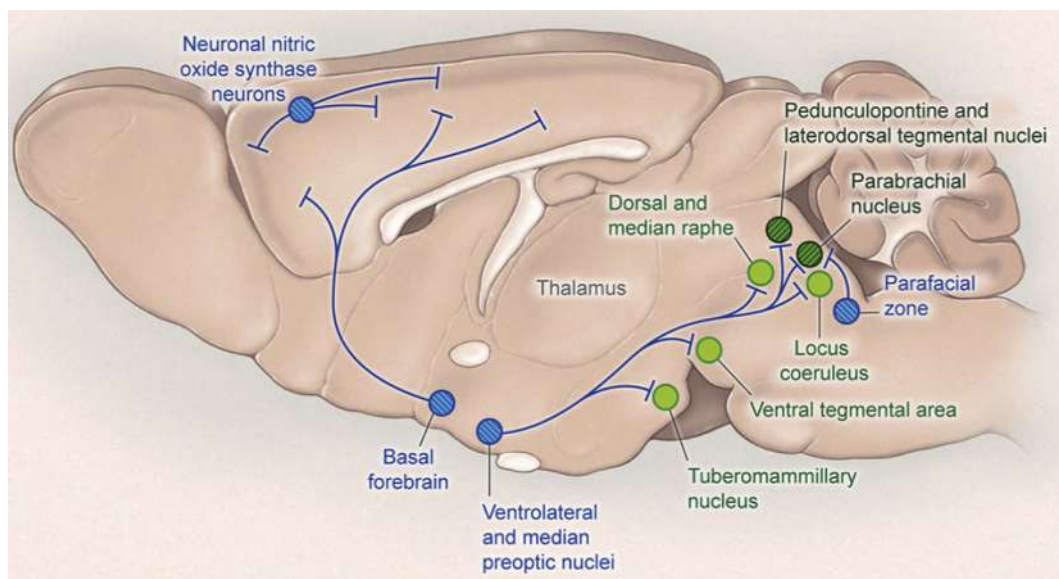


Fig. 9.2 Neurobiology of slow-wave sleep. GABA-ergic neurons in the ventrolateral preoptic area and median preoptic nucleus in the hypothalamus (shown here in rodent brain) promote sleep by inhibiting wake-promoting neurons in the caudal hypothalamus and brainstem. These hypothalamic nuclei are activated by general anesthetics. (Redrawn from Scammell TE, Arrigoni E, Lipton JO. Neural circuitry of wakefulness and sleep. *Neuron*. 2017;93[4]:747–765.)

SUBCORTICAL NUCLEI REGULATING AROUSAL

It was hypothesized in the mid-1990s that anesthetics suppress consciousness by actions at the subcortical nuclei that evolved to control sleep-wake cycles.¹¹ The past decades have supported the hypothesis that anesthetics interact with a number of these sleep-wake centers,^{12,13} although precise interactions and contributions to the state of general anesthesia have yet to be elucidated. The following is a description of select subcortical nuclei in the brainstem and hypothalamus that mediate sleep-wake cycles¹⁴ and, potentially, some traits of anesthesia.

Brainstem

Locus ceruleus. Norepinephrine is synthesized in the locus ceruleus (LC), which is located in the pons and projects widely throughout the cortex.¹⁵ Like other monoaminergic neuronal populations, LC activity is highest during waking consciousness, decreased during nonrapid eye movement (NREM) sleep, and at its nadir during rapid eye movement (REM) sleep.^{16,17} Thus LC is associated with cortical arousal only during wakefulness and not with the cortical activation during REM sleep. LC neurons are hyperpolarized by halothane.¹⁸ The role of norepinephrine (generated by LC)

in anesthesia is further supported by studies demonstrating that barbiturate anesthesia time is increased by antagonizing norepinephrine and reduced by agonizing it.^{19,20} Norepinephrine transmission in the basal forebrain may be of particular relevance to anesthetic depth.²¹ It has been found that LC noradrenergic neurons modulate the state of isoflurane anesthesia as well as emergence therefrom.²² Of note, administration of ketamine is associated with an increase of activity in the LC²³ and appears to contribute to its anesthetic effects.²⁴

LC and the role of norepinephrine in hypnosis are of particular interest due to the role of the α -2 agonist dexmedetomidine in clinical care. Microinjection of dexmedetomidine in the LC results in reduced levels of consciousness that can be prevented by coadministration of the α -2 antagonist atipamezole.²⁵ After exposure to dexmedetomidine, brain changes somewhat mimic NREM sleep in that the LC and tuberomammillary nucleus (TMN) are deactivated, whereas the ventrolateral preoptic nucleus (VLPO) is activated.²⁶ Data in dopamine- β -hydroxylase knockout mice (which lack the ability to synthesize norepinephrine) demonstrate a *hypersensitivity* to dexmedetomidine, suggesting alternative mechanisms of action.²⁷ However, selective knockdown of α -2A adrenergic receptors from LC prevent dexmedetomidine-induced loss of righting reflex,²⁸ a marker of general anesthesia in rodents.

Laterodorsal/pedunclopontine tegmentum. Along with the basal forebrain, the laterodorsal tegmentum (LDT) and pedunclopontine tegmentum (PPT) in the pons are the brain's source of acetylcholine.²⁹ There are direct projections to the thalamus from LDT/PPT with a known role in the generation of slow oscillations and sleep spindles,³⁰ which together represent a neurophysiologic sign that information transfer to the cortex is likely blocked.³¹ As with the noradrenergic LC, activity of the LDT/PPT is high during waking consciousness and decreases during NREM sleep.¹⁵ However, in contrast to the LC and other monoaminergic neurons, the cholinergic LDT/PPT is also active during REM sleep, during which the cortex is aroused. Furthermore, activation of cholinergic neurons in LDT or PPT induces REM sleep.³² Thus, both states of cortical activation across the sleep-wake cycle are associated with high cholinergic tone. General anesthetics modulate cholinergic projections from the LDT/PPT. Sleep spindles occur during halothane anesthesia and are associated with decreased cholinergic transmission to the medial pontine reticular formation (PRF).^{33,34} There is evidence that synaptic and extrasynaptic γ -aminobutyric acid (GABA) receptors play a role in modulating LDT neurons,³⁵ which could provide a direct link to molecular mechanisms of numerous general anesthetics. However, there has been relatively little study of the role of LDT/PPT in anesthetic mechanisms, with a greater focus on cholinergic neurons in the basal forebrain.

Pontine reticular formation. The PRF is part of the reticular activating system, which plays an important role in cortical arousal. Although GABA is the primary inhibitory neurotransmitter in the brain, the actions of GABA in the PRF are associated with cortical arousal.³⁶ For example, there is increased time spent in the waking state when the GABA_A receptor agonist muscimol is microinjected in the

PRF.³⁷ When the GABA_A antagonist bicuculline is microinjected, wakefulness is suppressed, but REM sleep (another state of cortical arousal) is triggered. Vanini and colleagues found that *decreased* levels of GABA in the PRF correlated with isoflurane-induced unconsciousness, muscular hypotonia, and decreased respiratory rate.³⁸ Since the effects of anesthetics are normally associated with a *potentiation* of GABA activity, these findings highlight that a specific neuroanatomic and neurochemical milieu can play a unique and unexpected role in the mechanisms of consciousness and anesthesia. In addition, Vanini and colleagues found that GABA-ergic transmission in the rat PRF modulates the loss of consciousness induced by isoflurane but does not appear to affect emergence,³⁹ providing evidence for asymmetry between the two processes.

The mesopontine tegmental anesthesia area is located in the PRF. When pentobarbital is microinjected in this area, a reversible state with anesthetic traits is induced.⁴⁰ More recently, this phenomenon has been defined with greater spatial resolution, with the identification of around 1900 neurons in this area that can induce general anesthesia.⁴¹

Ventral tegmental area. Dopaminergic neurons of the ventral tegmental area (VTA) in the midbrain have not classically been considered key mediators of the sleep-wake cycle because of relatively less evidence of state-dependent changes compared with neurons in other brainstem nuclei. This view has been challenged in sleep neurobiology.⁴² A dopaminergic pathway regulating sleep-wake states has been identified in *Drosophila*⁴³ and dopaminergic neurons of the VTA have more recently been found to play a role in mammalian sleep.⁴⁴ There has been renewed interest in the ability of dopaminergic activity to reverse or accelerate recovery from general anesthesia. Studies of the dopamine agonist methylphenidate have revealed an ability to reverse the sedative effects of both isoflurane and propofol.^{45,46} VTA appears to be the source of the dopaminergic transmission mediating arousal during exposure to anesthesia, as evidenced by the fact that electrical stimulation of the VTA or selective stimulation of VTA dopaminergic neurons can reverse anesthetic-induced unconsciousness.^{47,48}

Hypothalamus

Ventrolateral preoptic nucleus. The anterior hypothalamus has long been hypothesized to play a role in sleep-wake regulation.⁴⁹ VLPO is a structure in this region that transmits GABA and galanin.⁵⁰ Neurons in VLPO are maximally active during NREM and REM sleep;^{51,52} the median preoptic nucleus (MnPO) is also active during sleep. Of note, the activity profile of GABA-ergic neurons in VLPO correlates with sleep amount, whereas the activity of GABA-ergic neurons in MnPO correlates with homeostatic sleep pressure or propensity.⁵³ Importantly, activity of the VLPO during sleep is correlated with inhibition of other arousal centers in the brainstem and hypothalamus.^{51,54} Given its potentially central role as a mediator of sleep, VLPO became an attractive candidate as a mediator of anesthetic-induced unconsciousness. Nelson and colleagues demonstrated activation of VLPO after systemic administration of propofol or pentothal⁵⁵; recent studies have examined the mechanistic significance of these findings. Eikermann and colleagues conducted studies of rats with *chronic* lesions of

VLPO, finding that ablation of VLPO resulted in sleep deprivation (as expected) but conferred *increased sensitivity* to the effects of isoflurane.⁵⁶ This finding would argue against a critical role of VLPO in the mechanism of anesthesia. However, Moore and colleagues demonstrated that *acute* lesions of VLPO conferred *resistance* to the effects of isoflurane, an effect that appeared to be mediated specifically through the sleep-active neurons in VLPO.⁵⁷ These neurons are actually depolarized (i.e., activated) by isoflurane. Taken together, these data suggest that VLPO plays a role in anesthetic-induced unconsciousness (as evidenced by acute lesion data), but that the effects of sleep deprivation associated with chronic VLPO lesions could overwhelm this role. Curiously, direct administration of dexmedetomidine, an α -2-adrenergic agonist, to the VLPO can destabilize the state of isoflurane anesthesia.⁵⁸

Orexinergic neurons. Orexinergic neurons are found in the lateral hypothalamus and provide an important arousal stimulus for the cortex. There are two types of orexin (A and B), which are also referred to as hypocretins. Orexinergic neurons innervate other arousal centers in the brainstem and basal forebrain, and fire maximally in the waking state, are suppressed during NREM sleep, and show occasional bursts during phasic REM sleep.^{59,60} Dysfunction of the orexinergic system is associated with narcolepsy in both human and animal models.^{61,62} The often dramatic delay of anesthetic emergence in narcoleptic patients⁶³ has motivated the study of orexin in anesthetic mechanism. Orexins attenuate the effects of isoflurane,⁶⁴ propofol,⁶⁵ ketamine,⁶⁶ and barbiturates,⁶⁷ using various measures. Local infusion of orexin in the basal forebrain is associated with electroencephalographic (EEG) arousal and decreased emergence time in animals anesthetized with sevoflurane⁶⁸ and isoflurane.⁶⁹ Microinjection of propofol in the perifornical region of the hypothalamus (the locus of orexinergic neurons) is associated with a decrease in cortical acetylcholine, an important mediator of arousal.⁷⁰ Importantly, both genetic and pharmacologic studies have demonstrated that orexins play an important role in emergence from sevoflurane and isoflurane anesthesia, but not induction.⁷¹ This seminal study suggested that there is a distinct neurobiology of induction and emergence, and formed the basis for a theory of “neural inertia” across state transitions.⁷² Of note, halothane did not show an effect on orexinergic neurons and emergence time was not altered in orexin knockout mice.⁷³ These findings have been confirmed for propofol anesthesia: propofol reduces activity in orexinergic neurons in rats and infusion of orexin in the basal forebrain affects emergence time, but not induction time.⁷⁴ Orexins might facilitate emergence from anesthesia through actions in the basal forebrain.⁷⁵

Tuberomammillary nucleus. The TMN is located in the caudal hypothalamus and is the brain’s source of histamine, an arousal-promoting transmitter. TMN activity and histamine levels are highest during wakefulness and lowest during sleep;⁷⁶ the TMN is thought to have a relationship of reciprocal inhibition with the sleep-promoting GABAergic neurons of the VLPO.^{51,54,77} Histamine release in the anterior hypothalamus is depressed during sleep⁷⁸ and halothane anesthesia.⁷⁹ Systemic administration of propofol,

pentothal, and the GABA agonist muscimol all result in decreased activity in the TMN.⁵⁵ Microinjection of histamine in the nucleus basalis magnocellularis of the basal forebrain reverses the depressant effects of isoflurane on the EEG, an effect likely mediated by H1 histamine receptors.⁸⁰ A recent study in which GABA_A receptors were genetically removed demonstrated that histaminergic neurons are resistant to the effects of propofol.⁸¹ However, at the behavioral level, there was no effect of propofol on loss-of-righting reflex, a surrogate for anesthetic-induced unconsciousness. Thus the role of TMN and histaminergic transmission in the mechanism of anesthesia is still unclear.

ROLE OF THE THALAMUS

The thalamus is composed of more than 50 nuclei and subnuclei that can be grossly classified as relays for sensory input from the periphery or multimodal, integrative regions that receive input from the cortex. Additionally, the thalamus is critical for transmitting arousal signals from the brainstem and for regulating cortical communication. The involvement of the thalamus in arousal, sensory processing, and cortical computation is likely critical for normal consciousness. As such, the thalamus has been of continued interest to those investigating mechanisms of anesthetic-induced unconsciousness.

The thalamus has been proposed as an ON/OFF switch for anesthetic state transitions.⁸² This theory was generated based on the consistent metabolic depression of the thalamus by a number of inhaled and intravenous anesthetics⁸³⁻⁸⁵ (with the exception of ketamine),⁸⁶ suggesting that it could serve as an effective OFF switch. The hyperpolarization of the thalamus would shift tonic firing to burst firing that—as with sleep—would prevent afferent sensory stimuli from arousing the cortex. Importantly, however, sensory nuclei in the thalamus (and their connectivity to the cortex) appear to be less involved in anesthetic-induced unconsciousness compared to the higher-order or “nonspecific” nuclei.^{87,88}

Evidence for the thalamus as an ON switch has been derived primarily from animal experiments, in which stimulation of the centromedial thalamus by either nicotine or antibodies blocking voltage-gated potassium channels could reverse the effects of inhaled anesthetics.^{89,90} Although microinjection of large doses of nicotine into the thalamus could precipitate anesthetic emergence, antagonism of nicotinic acetylcholine receptors in the same location did not appear to contribute to anesthetic-induced unconsciousness. Central thalamic activation results in behavioral improvement in humans with traumatic brain injury.⁹¹ A study in humans demonstrated that (spontaneous) activation of the thalamus along with other subcortical structures is correlated with recovery from anesthesia, suggesting the involvement of the thalamus in the primitive or “core” consciousness observed at emergence.⁹²

Nonspecific nuclei of the thalamus have been proposed as a computational blackboard for the cortex.⁹³ Thus if the mechanism of anesthetic-induced unconsciousness was achieved primarily by a suppression of cortical computation, a depressed thalamus should be the result. To address this question, Velly and colleagues conducted a neurophysiologic study using scalp EEG (reflecting cortical signals)

and subthalamic nuclei electrodes (argued to reflect thalamic activity).⁹⁴ Induction of anesthesia with either propofol or sevoflurane was associated with cortical rather than subcortical changes, suggesting that the depression of the thalamus identified through neuroimaging studies reflected an effect rather than a cause of anesthetic-induced unconsciousness. Other case studies in humans suggest concurrent suppression of thalamus and cortex during induction of propofol.⁹⁵ Furthermore, more precise neurophysiologic recordings of thalamus and cortex in animal models suggest that effects of propofol on central thalamic activity precede effects on the cortex; interestingly this was found during the spontaneous induction of sleep as well.⁹⁶ Other animal studies have found that attenuation of high-frequency oscillations by propofol is more pronounced in the thalamus than the cortex.⁹⁷

The prior two possibilities (thalamic switch or computational blackboard for the cortex) treat the thalamus as a passive player in general anesthesia. However, more recent data suggest that it might, instead, play an active role. A computational study using human EEG data and modeling suggested that the action of propofol on GABA receptors in the nucleus reticularis generates a hypersynchronous alpha rhythm (8–13 Hz) with the frontal cortex that blocks sensory input.⁹⁸ Hypersynchrony of alpha may block the flexible corticocortical communication required for normal consciousness.⁹⁹ A recent animal study identified alpha synchronization between thalamus and medial prefrontal cortex during propofol induction.¹⁰⁰ The potential role of thalamocortical interactions in anesthetic-induced unconsciousness prompts further discussion of the thalamus and its connectivity to the cortex.

CORTICAL-SUBCORTICAL CONNECTIVITY

The closely integrated function of the cortex and thalamus suggests that the two can be treated as a single *thalamocortical system*. The thalamocortical system undergoes state-dependent changes across the sleep-wake cycle, and is thought to play a critical role in consciousness. This role is defined by its ability to integrate the activities of functionally diverse cognitive modules, a property that is critical for subjective experience.¹⁰¹

Recent studies using functional magnetic resonance imaging (fMRI) have refined the role of thalamocortical connectivity in anesthesia. One study identified a propofol-induced disruption of connectivity between the thalamus and lateral frontal-parietal networks.¹⁰² Similarly, a study of the specific nuclei (linked to particular sensory modalities) and nonspecific nuclei (linked to integrative functions) found that disrupted connectivity between the nonspecific nuclei and the cortex best accounted for a reduction in the level of consciousness by propofol.⁸⁷ Recently, the inhaled anesthetic sevoflurane has been shown to functionally disconnect the thalamus and cortex, especially frontal cortex.^{103,104} Notably, a consistent finding of neuroimaging studies is that the thalamocortical connectivity of primary sensory networks is relatively well preserved despite anesthetic-induced unconsciousness.

The finding of impaired thalamocortical connectivity in association with anesthetic-induced unconsciousness has not been universal. An fMRI study of propofol revealed more

profound functional disconnections between the cortex and putamen, a subcortical structure in the basal ganglia.¹⁰⁵ In contrast, thalamic connectivity was relatively well preserved. The potential role of the striatum (composed of the putamen and caudate) in anesthetic-induced unconsciousness has been demonstrated in a study of rats undergoing isoflurane anesthesia.¹⁰⁶ This study was conducted with fMRI and found that the functional connection between the frontal cortex and the basal ganglia was disrupted during general anesthesia. A functional disconnection of association cortex and subcortical structures has also been shown by fMRI in a study of propofol-induced unconsciousness in humans.¹⁰⁷ With improved spatial resolution of 7T fMRI machines and templates for brainstem nuclei,¹⁰⁸ future studies can focus on identifying with greater precision the critical subcortical and cortical interactions or functional disconnections that contribute to anesthetic-induced unconsciousness.¹⁰⁹

CORTICAL CONNECTIVITY AND DYNAMICS

The last three sections were organized according to a bottom-up approach to consciousness and anesthesia, starting with the brainstem, and then moving to the diencephalon and thalamocortical system. Sleep is clearly generated through such bottom-up mechanisms¹⁴; however, anesthetics may suppress the *level* of consciousness through bottom-up mechanisms and *content* of consciousness through top-down (i.e., cortical) mechanisms.¹²

Early studies using positron emission tomography (PET) demonstrated regional depression in cortical areas, including lateral and medial frontal-parietal networks.¹¹⁰ Disruption of fMRI-based functional connectivity in frontal-parietal networks has been demonstrated during anesthetic-induced unconsciousness induced by drugs with distinct molecular mechanisms, including propofol,^{102,111} sevoflurane,¹⁰³ and ketamine.¹¹² However, the cortex is amenable not only to fMRI but also neurophysiological techniques of assessing connectivity, enabling data on anesthetic-induced unconsciousness with improved temporal resolution. EEG can be used to measure functional connectivity (the statistical covariation of the activities of brain regions), and directional or effective connectivity (the presumed causal influence of one brain region on another).¹¹³ Using such techniques applied to EEG (in some cases with concomitant fMRI), disruption of frontal-parietal connectivity and surrogates of information exchange in humans has been consistently observed after administration of a variety of anesthetics with diverse molecular targets.^{104,114–117} It has been suggested that such disruption of frontal-parietal connectivity, with implications for impaired information transfer of relevance to consciousness, might be a common mediator and proximate cause of anesthetic-induced unconsciousness.¹¹⁸

The inhibition of frontal-parietal communication is likely representative of a more global disruption of cortical communication. A study using high-density EEG and transcranial magnetic stimulation revealed an inhibition of cortical-effective connectivity after midazolam-induced unconsciousness.¹¹⁹ After administration of the benzodiazepine, local cortical activation could be observed at the site of magnetic stimulation, but robust evoked potentials

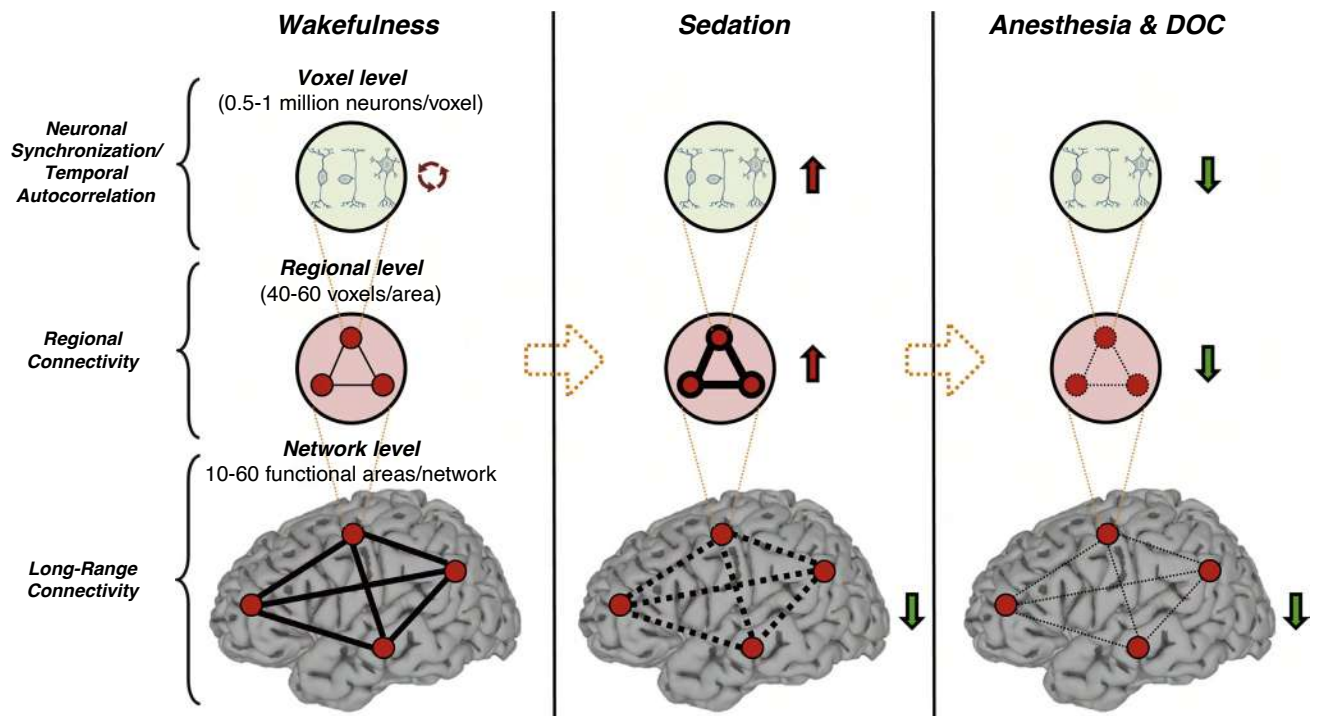


Fig. 9.3 Schematic summary of the consecutive stages of unconsciousness. Relative to the control state of wakefulness (*left column*), sedation (*middle panel*) is marked by an increase of local/regional signal synchrony and consequent breakdown of global connectivity. Deep surgical anesthesia or disorders of consciousness (*DOC*, *right column*) is associated with collapse of both local/regional synchrony and global connectivity. (From Huang Z, Liu X, Mashour GA, Hudetz AG. Timescales of intrinsic BOLD signal dynamics and functional connectivity in pharmacologic and neuropathologic states of unconsciousness. *J Neurosci*. 2018;38[9]:2304–2317.)

were terminated at less than 100 milliseconds, and cortical communication was limited. Of note, this finding is consistent with findings in NREM sleep.¹²⁰ This perturbational approach determined that the complexity of cortical response to a stimulus decreased during sleep, general anesthesia, and disorders of consciousness.¹²¹ The concordant findings across physiologic (sleep) and pharmacologic (anesthesia) states of unconsciousness may reflect a common neurophysiologic mechanism of disrupted cortical connectivity through slow oscillations, which share a number of characteristics in NREM sleep and general anesthesia.¹²² A study of three epilepsy patients with intracranial neurophysiologic monitoring¹²³ revealed that, within 5 seconds of propofol-induced unconsciousness, there was a dramatic increase in the power of slow oscillations. Although single-unit neuronal activity was initially suppressed, it returned to baseline (or above baseline) but was fragmented into highly active and quiescent periods. Neural firing became coupled with the slow oscillation. However, the slow oscillations themselves demonstrated decay in phase coupling with increased distance across the cortex. Thus neuronal spike activity became fragmented into “on” and “off” periods, which became temporally uncoordinated across the cortex. These neurophysiologic conditions dramatically reduce the probability of meaningful corticocortical communication.

More recent trends of analyzing cortical changes during states of unconsciousness take a dynamic approach that reflects not just connectivity configurations but the repertoire of states that can be accessed during general anesthesia. For example, there is a contraction of dynamic repertoire and neural signal diversity during

propofol-induced unconsciousness^{124,125} that would preclude the kind of flexibility required for normal conscious experience. Dynamic patterns are impaired during general anesthesia and cortical dynamics are stabilized during general anesthesia.¹²⁶ This is likely a discrete, multistage process with distinct dynamic signatures during sedation and general anesthesia (Fig. 9.3).^{127,128} Relating dynamics to connectivity, it appears that the repertoire of functional connectivity patterns in the primate brain becomes more tethered to structural/anatomical connectivity patterns during the anesthetized state.¹²⁹ Of note, animal studies suggest that the return of consciousness after general anesthesia is defined by discrete neural states that reconfigure during emergence.¹³⁰

In the next section, we discuss memory, the thread that links conscious experiences together to form the narrative of “self.”

Memory

HISTORY AND TERMINOLOGY

Modern understanding of the structure, function, and organization of human memory is deeply informed by the study of amnesia. The most renowned demonstration of this principle occurred in 1957, when Brenda Milner reported the remarkable case of Henry Gustav Molaison (1926–2008),¹³¹ an amnesiac who would become known famously as H.M., and who would represent the single most influential case study in the history of neuroscience. In a procedure intended to treat a refractory seizure disorder,

significant portions of the medial temporal lobe (MTL) were removed bilaterally—including the hippocampus, amygdala, and adjacent parahippocampal gyrus. H.M. developed profound and enduring anterograde amnesia, and was unable to establish any new conscious memory, irrespective of the sensory modality. He also developed a temporally graded window of retrograde amnesia, with impaired recall of events occurring during the 3 years preceding his surgery. However, most of his associated cognitive functions—perceptual processing, language, attention, access to semantic knowledge, and capacity to retain small packages of information in constant rehearsal—remained largely or entirely intact. Prior to this report, the prevailing theory—articulated by Canadian neuropsychologist Donald Hebb¹³²—was that there was no brain region dedicated to memory function. Instead, memory processes were thought to be distributed and integrated into region-specific perceptual and cognitive functions. For example, the visual attribute of a memory would be wholly served within the striate and extrastriate cortical regions responsible for visual perception. The description of H.M. immediately disproved this model. It became clear that the MTL was a specialized and obligatory structure for the establishment and early maintenance of *all* conscious memory, irrespective of the modality. The trajectory of memory research was profoundly transformed. Initially, largely independent branches evolved to focus on the structural-functional organization of the MTL (Fig. 9.4A) and the nature of cellular-level neuroplastic processes—the latter notably marked by the description of long-term potentiation (LTP) by Timothy Bliss and Terje Lømo in 1973.¹³³ Subsequently, emerging technologies enabled the development and investigation of systems-level constructs. Examples include the use of EEG and magnetoencephalography to assess the role of oscillatory phase synchronization in neuronal assemblies,¹³⁴ fMRI to identify large-scale networks associated with specific memory functions,¹³⁵ and machine learning to classify complex neural network patterns predictive of memory.¹³⁶

Amnesia is the term used to describe one of the cardinal properties of general anesthesia. As understood by most anesthesiologists and laypeople, this description is phenomenological; it states that patients do not recall the events that occur to them while receiving anesthesia. However, this usage confuses a critical mechanistic and semantic distinction. Patients in the deepest states of anesthesia are unable to process and bind perceptual elements into an integrated conscious experience. From the perspective of cognitive neuroscience, the “amnesia” of general anesthesia is not a primary failure of memory, but rather a failure of consciousness. It simply reflects that a conscious experience cannot be reconstructed by memory processes when it does not exist in the first place. Further confusion is added by the frequent use of the term *awareness*—a synonym for conscious perception—to describe the case in which a patient consciously recalls events occurring during the administration of an anesthetic. This ignores the fundamental principle that memory is functionally dissociable from consciousness. Awareness is *necessary* for the establishment of memory under anesthesia, but it is not *sufficient*. Conscious recall can occur only if awareness is accompanied by memory processes in the MTL and elsewhere that establish and preserve a representation that can be reconstructed later.

These important distinctions establish the axiom that patients who form memories while under anesthesia cannot have been truly unconscious; they must have possessed some conscious substrate from which the memory derives. However, the converse inference is not always true: the existence of consciousness will not necessarily lead to the existence of memory if an anesthetic drug is present. Evidence to support this statement is unambiguously encountered in everyday anesthetic practice—patients receiving a small dose of propofol or midazolam who engage in a cogent conversation that they are later unable to recall, or in patients emerging from general anesthesia who follow commands to demonstrate that extubation can proceed safely, yet later cannot recall anything related to this clearly conscious event. Anesthetic drugs must therefore have direct effects on memory processes that are dissociable from those on consciousness—and it is this observation that provides a framework for the systematic study of how anesthetic drugs affect memory.

Organization and Function of Normal Memory

MULTIPLE MEMORY SYSTEMS

When the term *memory* is used in everyday language, it almost always refers to declarative memory. Declarative memory is the representation of prior events and knowledge that is accessible to consciousness and can be manipulated by attention and executive function. It is the form of memory referred to in the context of anesthetic amnesia.

Further important organizational structure exists within declarative memory. The first is the distinction between episodic and semantic memory. *Episodic memory* is the recollection of events with a clear spatiotemporal context (as when recalling autobiographical events with a distinct sense of personal experience, time, and place), whereas *semantic memory* is the capacity to recall and apply meaning, facts, and knowledge without spatiotemporal context (as when recalling that Mount Everest is the tallest mountain in the world without any sense of time and place for the acquisition of that knowledge). The episodic memory system is fast-mapping and highly dependent on the MTL as well as frontal and parietal structures,^{137,138} whereas the semantic memory system is slower and involves distributed cortical regions that closely map to the default mode network, a large-scale system that is active during the resting state and flow of spontaneous cognition.^{135,139} The second organizational structure within episodic memory is the distinction between recollection and familiarity. *Recollection* involves remembering specific qualitative contextual details about a prior event, whereas in a *familiarity* judgment, there is a sense that an item has been encountered previously, but beyond that there are no added contextual details. The accepted understanding is that recollection and familiarity arise from distinct processes and neural architecture within the MTL. The perirhinal cortex receives input from sensory association areas and supports familiarity judgments through encoding and retrieval of the identifying qualities of an item (the “what” information). The parahippocampal cortex and entorhinal areas receive input from areas

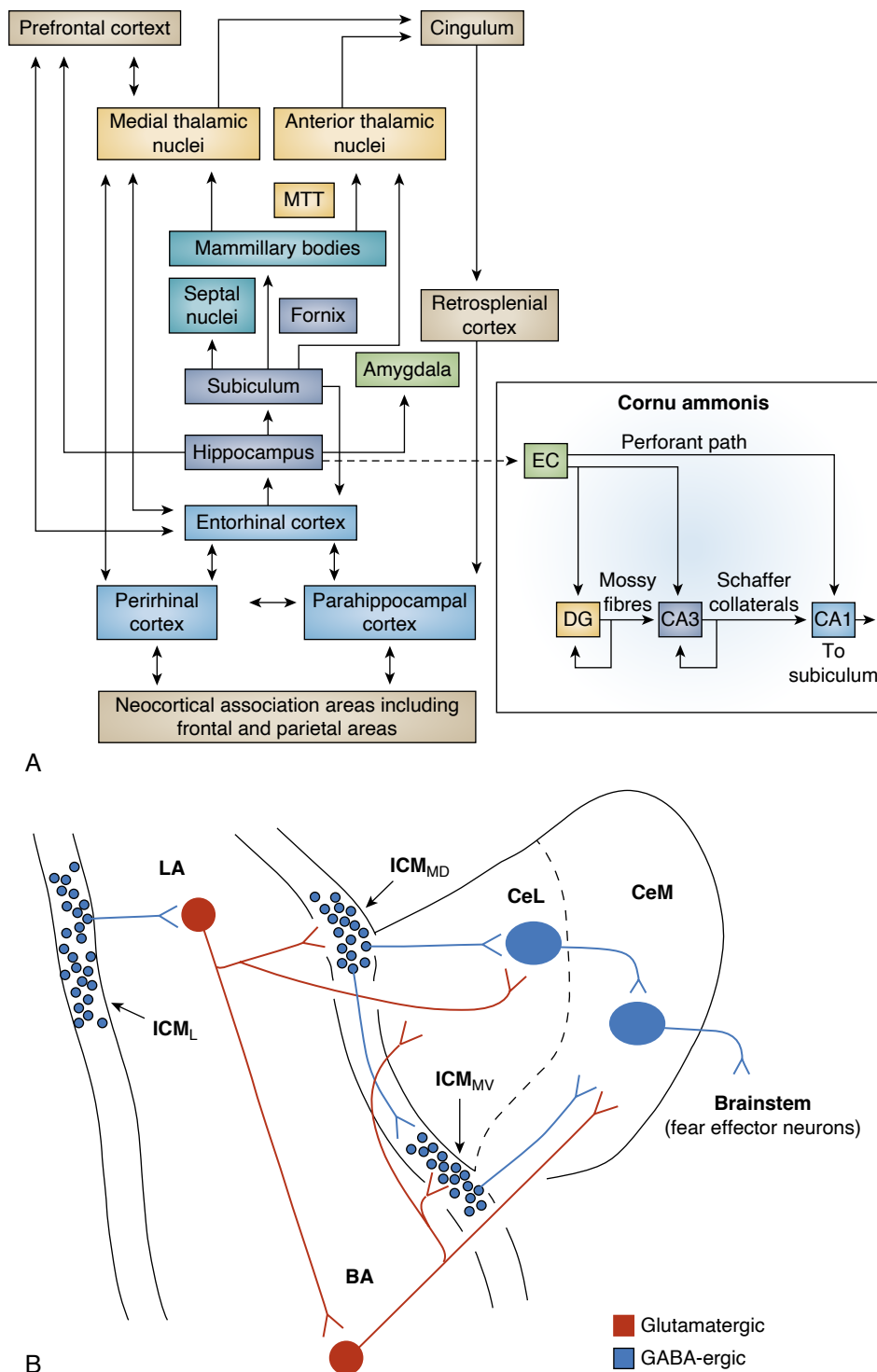


Fig. 9.4 Memory systems in the medial temporal lobe. (A) Representation of unidirectional and bidirectional network connectivity of structures in the medial temporal lobe, diencephalic nuclei, and neocortical association areas. *CA*, Cornu ammonis; *DG*, dentate gyrus; *EC*, entorhinal cortex; *MTT*, mamillothalamic tract. (B) Intrinsic connectivity of the amygdala. *BA*, Basal nucleus; *CeL*, central nucleus lateral segment; *CeM*, central nucleus medial segment; *ICM_L*, lateral intercalated cell mass; *ICM_{MD}*, dorsomedial intercalated cell mass; *ICM_{MV}*, ventromedial intercalated cell mass; *LA*, lateral nucleus. ([A] Modified from Bartsch T, Butler C. Transient amnesic syndromes. *Nat Rev Neurol*. 2013;9[2]:86–97, Figure 2. [B] Modified from Duvarci S, Pare D. Amygdala microcircuits controlling learned fear. *Neuron*. 2014;82[5]:966–980, Figure 2.)

processing spatial information (the “where” information), and support recollection through encoding and retrieval of context. The hippocampus links these two, binding item and context information, and appears necessary for recollection, but plays little or no role in familiarity.^{140,141}

Other forms of memory are dissociable from declarative memory. Findings that amnesiacs could learn a hand-eye coordination skill even while possessing no memory of the task led to the distinction between declarative and *procedural* memory, which is dependent on the caudate nucleus.

Subsequently, memory-impaired patients were also found to have intact *priming*,¹⁴² which is a nonconscious process in which exposure to a stimulus influences the response to a later stimulus—for example, amnesiacs can name pictures 100 milliseconds faster if they have seen them previously, despite having no declarative memory of the exposure.¹⁴³ The neural bases for priming effects are thought to arise within regions serving perceptual and contextual processing of the stimulus, and so vary with the nature of the task.¹⁴⁴ Lastly, a large body of work has extensively elucidated the *emotional memory* system. The classic experimental model is Pavlovian fear conditioning and its variants, wherein an emotionally neutral conditioned stimulus is paired with an aversive unconditioned stimulus, leading to an involuntary associative physiologic and/or behavioral response to the conditioned stimulus. The circuitry involves convergence of afferent information from all sensory modalities on the lateral nucleus of the amygdala, and the basolateral and central nuclei projecting widely to modulate processes in cortical and subcortical regions (see Fig. 9, 4B).^{145,146} With the progressive description of these and other functional distinctions, the framework for understanding memory eventually shifted to models that contained multiple memory systems in the brain, distributed functionally and anatomically.¹⁴⁷

Working memory refers to the capacity to maintain limited amounts of information in the stream of consciousness, which can be manipulated to perform complex cognitive tasks such as reasoning, comprehension, and learning.¹⁴⁸ The concept evolved from, and has largely replaced, earlier ideas about *short-term memory*, but the terms should not be used interchangeably, as working memory implies both a short-term memory store and the capacity for manipulation. The most influential current model, first proposed by Baddeley and Hitch in 1974,¹⁴⁹ divides working memory into capacity-limited component subsystems: a *phonological loop* that maintains information through vocal or subvocal rehearsal, such as when one holds a telephone number in mind; a *visuospatial sketchpad*, which holds and manipulates spatial, visual, and kinesthetic information; and a *central executive*, which is responsible for regulating selective attention and inhibition. A fourth subsystem, the *episodic buffer*, was later added to the model¹⁵⁰ and is responsible for the temporary storage of multidimensional representations and integration with declarative memory.

Working memory has long been understood to not involve the MTL,¹⁵¹ although the belief that it has no role has recently been questioned in studies of spatial working memory.¹⁵² Notwithstanding, current understanding of working memory suggests that it is served by persistent activity and flexible resource allocation in a distributed cortical network, with a critical hub in the dorsolateral prefrontal cortex (DLPFC) interconnected with parietal cortex, thalamus, caudate, and globus pallidus.¹⁵³ The functional and structural distinction between working memory and the MTL does not mean that working memory and declarative memory systems do not interact. Working memory depends on declarative memory representations to provide semantic meaning and context. During working memory tasks, cortical perceptual areas associated with representations of declarative memory become activated and show increased synchrony with prefrontal regions.¹⁵⁴ Reciprocally, encoding of declarative memory is strongly influenced by the nature of processing

occurring in working memory, with deeper levels of executive processing resulting in better learning.¹⁵⁵

The transition of memory from short-term stores through to stable long-term stores is experienced as continuous and was assumed to reflect sequential transfer across systems. However, this model is challenged by rare case examples of patients who have a selective short-term memory deficit but intact declarative memory function, and recent studies have reinforced the view that memories are formed in multiple systems in parallel.¹⁵⁶

LONG-TERM POTENTIATION, SYNAPTIC TAGGING, AND THE CONSOLIDATION MODEL OF MEMORY

The consolidation hypothesis of memory was first proposed by Müller and Pilzecker in 1900.¹⁵⁷ They noted that memory for new information could be disrupted by learning other information shortly after the initial training. This effect, called *retroactive interference*, is temporally graded such that the susceptibility of the memory is greatest immediately after learning and decreases with time. Müller and Pilzecker proposed that the memory trace must initially exist in a fragile state, but subsequently becomes stable through the process of *consolidation*. The consolidation hypothesis remains the framework for understanding the temporal course of memory processes and behavior.

For a memory trace to be consolidated, it must of course be created. The term used to describe this process is *encoding*. Encoding implies that the networks mediating the neural representation of an event as it is experienced do not immediately return to their previous state and are modified in such a way that potentiates reactivation of that representation. The *synaptic plasticity and memory hypothesis* states that activity-induced synaptic plasticity is both necessary and sufficient for the information storage underlying memory,¹⁵⁸ and within this framework encoding implies that some form of synaptic plasticity has been initiated. However, encoding cannot in itself assure the propagation of a memory trace. Encoding creates the potential for the formation of a long-term memory.

The minimal events that constitute the neural correlates of encoding are incompletely understood. Cellular models demonstrate that functional changes in synaptic strength can occur in the absence of any structural change in dendritic spines.¹⁵⁹ The perpetuation of these initial changes through structural and functional remodeling represents the neural correlate of memory consolidation. The prevailing cellular model for this is LTP,¹⁶⁰ which describes a durable increase in synaptic transmission efficiency following a stimulation protocol. It is now recognized that LTP occurs richly throughout the hippocampus, as well as in other afferent pathways.¹⁶¹ LTP can be induced by nonphysiologic high-frequency stimulation, but also by stimulation protocols that resemble physiologic activity, the most important being bursts in the theta range (4–8 Hz).¹⁶² This is of notable relevance to memory, as synchronized hippocampal theta oscillations appear critical to successful memory behaviors.¹³⁴

The breadth and depth of literature on the mechanisms of LTP are far too voluminous to summarize here. Nonetheless, certain principles are essential and relevant to anesthesia studies and can be stated succinctly. The induction of most forms of LTP requires activation of postsynaptic N-methyl-D-aspartate (NMDA) receptors,¹⁶³

followed by influx of Na^+ and Ca^{2+} . This rise in intracellular Ca^{2+} is the critical trigger for LTP. Calcium-calmodulin-dependent kinase II (CaMKII) is then activated and autophosphorylated,^{164,165} leading to cytoskeletal reconfiguration.¹⁶⁶ Activation of several other cell-signaling cascades also contribute to LTP. The terminal expression of LTP is protein synthesis, occurring in both the soma and local dendrites, and resulting in enduring structural changes at the synapse.¹⁶⁷ Protein synthesis inhibitors have been demonstrated to consistently prevent sustained LTP in vitro and learning in vivo.¹⁶⁸

LTP thus proceeds in two phases: early LTP (E-LTP) is independent of protein synthesis and can be sustained across an interval of minutes to a short number of hours, while late LTP (L-LTP) is dependent on intracellular signaling and protein synthesis and can be sustained across many days. The *synaptic tagging and capture hypothesis* provides a mechanistic explanation for the observation that the persistence of synaptic potentiation (and memory) is influenced by events surrounding encoding.¹⁶⁹ In this model, synapses activated during E-LTP become tagged via a protein synthesis-independent mechanism. This tag establishes the potential for sustained L-LTP, but for the cascade to continue the tags must capture plasticity-related proteins (PRPs) synthesized in the soma or dendrites in response to neural activity. The synaptic tagging model explains how the thousands of dendrites of a single neuron can be engaged in memory stabilization processes in various states of evolution, because the tagging and PRP capture need not occur as a singular event.

RECONSOLIDATION

A major shift in the understanding of consolidation occurred in 2000, when Nader et al. reported that an old memory for auditory fear conditioning, which would normally not be sensitive to protein inhibitors, can be made newly sensitive if it is retrieved.¹⁷⁰ The implication is that retrieval of a memory renders it transiently plastic, after which it restabilizes. This process is termed *reconsolidation*, and the period of plasticity is termed the *reconsolidation window*. The mechanism shares many of the LTP processes that are associated with initial consolidation, but also has features that are quite distinct at both the cellular and systems level.¹⁷¹ Reconsolidation serves as a modulatory process that enables the strengthening of an existing memory, but it is also clear that it provides a window within which an existing memory is malleable and can be updated with the addition of novel information, or diminished through interruption of restabilization processes.¹⁷² These latter properties have generated significant translational interest because of repeated demonstrations of the ability to modify or even eliminate fear memory in animals through pharmacologic and behavioral interventions that interrupt reconsolidation.¹⁷³ Indeed, the erasure of fear memory in humans can be demonstrated through performance of a behavioral extinction procedure during the reconsolidation window.¹⁷⁴

PHASE SYNCHRONIZATION AND COUPLING

The neurons in assemblies and networks undergo oscillatory activation and inhibition. Phase synchronization of

these oscillations supports neural communication by creating transient and dynamic associations between different functional brain regions. Phase synchronization appears to be fundamental to neural plasticity and memory,¹³⁴ and numerous studies have demonstrated that dynamics of synchrony during memory tasks can be related to both long-term¹⁷⁵ and working memory¹⁷⁶ performance. Gamma-band (30-100 Hz) synchrony is believed to support an important form of Hebbian plasticity, termed spike-timing-dependent plasticity, in the hippocampus. Both computational models and experimental data provide evidence that rapid changes in the frequency and phase of gamma rhythms modulate this form of plasticity through coordination of presynaptic spikes in assemblies of neurons, effectively classifying which assemblies are interpreted as a single event.¹⁷⁷

Another property of relevance to memory coding is the coupling of gamma oscillations to the phase of slower, synchronized oscillations within the theta-band (4-8 Hz).¹⁷⁸ Theta oscillations, which are prominent in the hippocampus and entorhinal cortex, undergo phase resetting in response to a stimulus. The phase reset involves widely distributed regions, and is thought to serve as an efficient mechanism for the optimization of interregional communication.¹⁷⁹ Theta phase resetting and synchrony in the hippocampal-entorhinal system are implicated in declarative memory,¹⁸⁰ and in the amygdala-hippocampal system in fear-based memory.¹⁸¹ Coupling between theta phase and gamma amplitude is connected to declarative memory formation,^{180,182} and, in an elegant model, is proposed to form a code for representing multiple items and spatial representations in an ordered way.¹⁸³ Coupling between theta phase and gamma phase enables a more precise temporal coordination of neuronal spikes, and may code for separated representations of multiple items in working memory.¹⁸⁴

Effects of Anesthetic Drugs on Declarative Memory Function in Humans

There are multiple potential pathways by which anesthetic drugs could interfere with memory and cause amnesia. Most of these cannot be directly assessed in humans, but the plausibility of candidate mechanisms can be indirectly examined by designing experiments that are informed by the characteristics of known memory processes. Studies of anesthetic effects on memory are most informative when conceptualized and designed with reference to robust methods and findings taken from the larger body of memory research.

BEHAVIORAL STUDIES OF RETROGRADE MEMORY EFFECTS

Notwithstanding some isolated case reports, systematic investigations have found no evidence that anesthetic drugs cause retrograde amnesia in humans. There is no retrograde amnesia associated with induction doses of thiopental,¹⁸⁵ methohexital,¹⁸⁵ or propofol.¹⁸⁶ Adult patients have normal memory for visual stimuli presented 4 minutes before administration of midazolam at doses of 2, 5, and 10 mg.¹⁸⁷ Similarly, memory is

normal for word lists learned in the preoperative holding area or operating room immediately before induction.¹⁸⁸ Studies in pediatric patients have shown normal memory for pictures presented immediately before sedation with midazolam,¹⁸⁹ propofol,¹⁹⁰ and dexmedetomidine.¹⁹¹ In controlled laboratory settings using human volunteers and target-controlled infusions, studies have found no impairment of memory for pictures¹⁹² or words^{193,194} presented before sedative infusions of propofol, midazolam, thiopental, or dexmedetomidine.

There is some evidence from controlled studies that anesthetic drugs may instead lead to the opposite effect, in which memory for material presented prior to drug administration is enhanced—a phenomenon termed *retrograde facilitation*. In one study using a mildly sedative dose of propofol,¹⁹⁵ memory for words learned immediately prior to drug administration was enhanced throughout the testing period of 24 hours. Similar effects are described for midazolam and other benzodiazepines in the psychopharmacology literature.¹⁹⁶ The mechanistic explanation for retrograde facilitation can be framed by an understanding of *retrograde interference*—the observation that mental exertion inhibits the consolidation of recently formed memories, with the newest memories being most vulnerable.¹⁹⁷ The induction of new LTP interferes with recently formed LTP and memory performance, even when the tasks are unrelated.¹⁹⁸ However, when the induction of new LTP is blocked by administration of a selective NMDA antagonist after learning, interference with recently formed LTP does not occur, and memory performance improves.^{199,200} A parsimonious explanation for the retrograde facilitation seen with propofol and benzodiazepines is that they similarly modulate the induction of new LTP via a GABA-ergic pathway, freeing consolidation resources that enhance survival of recently formed memory.

The findings of retrograde facilitation and absence of retrograde amnesia are suggestive that the key mechanisms of GABA-ergic anesthetic amnesia, at least in humans, involve very early processes in the consolidation cascade. It is more difficult to rationalize that downstream sequences represent principal targets. Were this the case, the expected effect would be interference with ongoing consolidation of memory for events occurring in the past, creating a window of retrograde amnesia. Nonetheless, an alternate model to the induction hypothesis suggests that GABA-ergic anesthetics exert a direct effect on downstream protein transcription processes,²⁰¹ based on evidence from rodent studies.²⁰²

The absence of pharmacologic retrograde amnesia does not dismiss the repeated finding that, in the clinical setting, a percentage of patients will not recall the immediate preanesthetic period—an effect that increases with age.²⁰³ Although not systematically studied, the complex noradrenergic-mediated effects of acute stress and anxiety on memory may be contributory.²⁰⁴

MATHEMATICAL MODELING OF ANESTHETIC AMNESIA

Mathematical modeling of memory decay has been used to characterize the amnesic effects of multiple intravenous anesthetic drugs,¹⁹³ which are accurately described by a two-parameter power decay function:

$$m_t = \lambda t^{-\psi}$$

where λ reflects the initial memory strength (an index of encoding), and ψ expresses the rate of decay (an index of consolidation). The drugs are characterized by marked differences in the way they modulate the two coefficients. Propofol is an archetypal amnesic drug—it permits robust encoding of material, but the information undergoes accelerated decay because of a failure of consolidation. In contrast, dexmedetomidine archetypally causes memory impairment because of a failure of information to be strongly encoded but has little or no effect on the subsequent consolidation of memories that are encoded. The benzodiazepine midazolam behaves like propofol at lower doses—selectively causing consolidation failure while leaving encoding intact—but with increasing dose a significant encoding impairment emerges. Thiopental in contrast causes marked encoding failure but demonstrates minimal effect on consolidation. The discrepant patterns observed imply that nonspecific GABA_A agonism is not per se sufficient to explain the ability of a drug to cause consolidation failure.

ANESTHETIC EFFECTS ON ATTENTION AND AROUSAL AS MODULATORS OF ENCODING

The effect of anesthetic drugs on encoding processes is related to the modulation of attention. Selective attention is governed by distinct networks mediating alertness, target orientation, and executive control²⁰⁵; involves modulation of neuronal phase synchrony across short- and long-range connections²⁰⁶; and is necessary for the successful establishment of declarative memory.²⁰⁷ The dominant effect of most anesthetic drugs on attention is decreased arousal, with the notable exception being the NMDA antagonist ketamine, which has dominant effects on orienting and selection.²⁰⁸ Thiopental mimics the effect of an attentional challenge task in decreasing activation in the left inferior prefrontal cortex (LIPFC), an effect that is not seen with propofol.²⁰⁹ When tested at 225 minutes, arousal is predictive of memory for thiopental and dexmedetomidine, while propofol causes a significantly greater loss of memory than would be predicted by modulation of arousal alone.¹⁹⁴ In mathematical modeling of anesthetic amnesia,¹⁹³ arousal predicts the coefficient of encoding strength with precision for a range of sedative concentrations of dexmedetomidine, thiopental, midazolam, and propofol. In sum, arousal is predictive of subsequent memory for drugs that impair memory dominantly through causing encoding failure.

NEUROIMAGING STUDIES OF CORTICAL ENCODING PROCESSES

A limited number of functional neuroimaging studies have evaluated the effect of anesthetics on cortical regional activation during memory encoding. An early PET study investigating sedative doses of propofol using a word memory task²¹⁰ identified that activation in LIPFC—a region associated with encoding and subsequent memory for language tasks—was conserved, suggesting that propofol did not block the processes required to support successful encoding. In contrast, activation in DLPFC—a region most associated with executive control functions and cognitive control of motor planning²¹¹—was decreased. In a subsequent investigation using an auditory depth of processing task, activation

of LIPFC was found to be decreased by sedative doses of thio-pental, but again was relatively unaffected by propofol.²⁰⁹ However, other studies have identified decreased activation in encoding areas. An fMRI study of sentence comprehension during light sedation with propofol found that decreased memory for sentences was related to decreased activation in the left inferior frontal gyrus (LIFG, within the LIPFC) and the middle temporal gyrus.²¹² Another study using a single-word encoding task showed decreased activation in LIFG at levels of propofol sedation causing memory loss, but also found intact connectivity between LIFG and a number of frontoparietal and temporal regions associated with verbal processing and memory tasks, including the middle temporal gyrus and precuneus.²¹³

Some functional neuroimaging studies have used experimental tasks similar to memory encoding tasks while evaluating sedative doses of anesthetic drugs, but have not assessed memory encoding as a primary aim; these are informative, but must be interpreted with caution. An early PET study of sedative doses of midazolam using a tone detection paradigm reported a dose-dependent activation decrease in Brodmann areas 9, 10, and 46, which overlap regions of both DLPFC and LIPFC.²¹⁴ An fMRI study of semantic word processing during propofol sedation found a dose-dependent decrease in activation of the LIFG, despite intact behavioral responses.²¹⁵ An fMRI study of low-dose dexmedetomidine reported a generalized suppression of bilateral prefrontal activation in Brodmann areas 9 and 10 during an emotional picture memory task, but encoding performance was not specifically analyzed.²¹⁶ A recent study using music stimuli found decreased activation in primary and secondary auditory processing areas at sedative doses of dexmedetomidine and midazolam, but not with propofol.²¹⁷ Other neuroimaging studies evaluating resting state networks, or passive activation in response to stimuli at nonresponsive levels of sedation, should not be viewed as memory experiments.

Neuroimaging Studies of Medial Temporal Lobe Function

Two methodologically similar studies used event-related fMRI to evaluate MTL activation during low-dose infusions of propofol and dexmedetomidine. The degree of amnesia caused by propofol can be linearly related to decreased activation in the hippocampus bilaterally,²¹⁸ an effect that behaviorally corresponded to a failure of consolidation processes. In contrast, dexmedetomidine does not reduce the overall level of hippocampal activity, but the subsequent-memory effect is attenuated, and dynamics of hippocampal activation are less predictive of subsequent memory²¹⁶; one interpretation is that this reflects a downstream effect from weakened cortical encoding processes. An earlier study of the benzodiazepine lorazepam and the cholinergic antagonist scopolamine found decreased memory to be correlated with decreased activation in the anterior hippocampus,²¹⁹ accompanied by decreased activation in the encoding-related regions in the fusiform gyrus and inferior frontal cortex. In a largely exploratory investigation, 0.25% sevoflurane was found to reduce hippocampal activation in response to auditory and visual stimuli, but no memory performance was assessed.²²⁰

A few studies have evaluated anesthetic effects using electrocorticography and depth electrodes implanted

in epileptic patients, but none have been dedicated to memory function. One notable study evaluated the effects of mildly sedative concentrations of propofol on hippocampal spectral coherence and power characteristics at rest.²²¹ The main finding was a significant increase in hippocampal-rhinal spontaneous coherence in the delta-band, but minimal changes in other bands; cortico-hippocampal coherence was not assessed.

STUDIES OF CORTICAL EVENT-RELATED POTENTIALS

A number of memory-dedicated studies have used the event-related potential (ERP)—a small but stereotypic positive and negative signal fluctuation in the EEG time-locked to a stimulus, and which can be isolated through summation over multiple (usually at least 50) identical trials. ERPs are thought to emerge not because new oscillations are induced, but because of a stimulus-induced phase resetting of ongoing oscillations.²²²

An early ERP study²²³ of scopolamine, lorazepam, and diphenhydramine suggested that drug-induced effects on arousal and memory may be electrophysiologically dissociable, although memory was not assessed directly. Changes in the P1N1 and N1P2 early complexes, associated with arousal, were seen for all three drugs. However, changes in later complexes associated with memory, notably the P3 and N2P3, were seen only with scopolamine and lorazepam, and not with diphenhydramine, an antihistamine that causes sedation but not amnesia. This work is extended by a series of studies evaluating intravenous anesthetics and using memory-specific experimental paradigms. In a verbal memory task, propofol amnesia was associated with decreased P300 amplitude at the time of the encoding task.²²⁴ In a subsequent study in which multiple drugs were carefully dosed to equivalent levels of sedation,²²⁵ the true amnestic drugs with memory effects independent of sedation—propofol and midazolam—caused decreased amplitude in the P300 and N2P3 components, with the latter being the best predictor of subsequent memory performance. Across all drugs studied, the N2 latency was related to reaction time, a surrogate measure of sedation.

Another study examined early- and mid-latency ERPs originating from the midline parietal precuneal region (Pz), and related them to coefficients describing the extent of encoding and consolidation failure.²²⁶ Across drugs from multiple classes and at multiple doses, consolidation failure was closely correlated with the P2 amplitude and N2 latency observed at the time of encoding. As the visual P2N2 complex is known to originate from synchronous theta oscillations,²²⁷ one possibility is that a common mechanism underlying the effect of anesthetic drugs on consolidation involves changes in theta oscillations across a distributed cortico-hippocampal network occurring at the time of consolidation induction. Further, the P2N2 and memory decay coefficient were also closely correlated with reaction time, which can be related to interregional synchrony.²²⁸ Unfortunately, no direct measures of cortico-hippocampal theta synchrony have yet been performed in studies of pharmacologic amnesia. Another analysis²²⁹ examined the *old-new effect*, a robust phenomenon in the parietal ERP that distinguishes the response to initial item

exposure from subsequent exposures as a marker of memory strength. Propofol and midazolam caused a significant decrease in the old-new effect at 27 seconds, even though memory performance was maintained, demonstrating again an early marker of impaired memory processes that precedes behavioral detection.

Studies using very different experimental paradigms have evaluated auditory ERPs at multiple concentrations of propofol.^{230,231} Propofol causes a dose-dependent decrease in mismatch negativity and early right anterior negativity, which are elicited in response to specific music signatures and are known to relate to associative memory operations. In contrast, the P1 complex, which derives from primary auditory processing, is not affected even at deep levels of sedation.

Nonhuman Studies of Anesthetic Effects on Memory Processes and Behavior in the Medial Temporal Lobe

ANESTHETIC EFFECTS ON MEDIAL TEMPORAL LOBE PLASTICITY

GABA-ergic interneurons project within and across subregions of the hippocampus,²³² providing an abundant density of possible targets for anesthetic drugs. In an early study using tetanic stimulation of the Schaffer collateral-commissural pathway,²³³ isoflurane blocked the induction of LTP, and long-term depression (LTD) by low-frequency stimulation. These effects were reversed by the addition of the GABA_A receptor antagonist picrotoxin, providing strong evidence that the effect of isoflurane on LTP is GABA-ergically mediated. In a similar protocol, amnestic concentrations of sevoflurane caused a failure of LTP which was prevented by the addition of the GABA_A antagonist bicuculline.²³⁴ A recent study of sevoflurane exposure in neonatal rats found reduced spine density of apical dendrites, synaptic ultrastructure damage, elevated expression of synaptic vesicle-associated proteins, and inhibition of LTP but not LTD.²³⁵ One study of isoflurane, in which the hippocampus was studied 24 hours following anesthetic exposure, somewhat unexpectedly found improvement in cognitive performance, accompanied by evidence of upregulation of the 2B subunit of the NMDA receptor and enhanced LTP.²³⁶

A series of investigations have demonstrated that propofol inhibits the induction, but not the maintenance, of LTP, and has no effect on LTD. The effect is blocked by the addition of picrotoxin, implicating a GABA_A receptor-mediated mechanism.^{237,238} Notably, one study found that propofol inhibited LTP only at anesthetic, and not at amnestic concentrations.²³⁸ Propofol has also been demonstrated to inhibit a number of component subprocesses of LTP.²³⁹⁻²⁴¹ Propofol reduces expression of activity-related cytoskeleton-associated protein (Arc) in the hippocampus in response to inhibitory avoidance training but does not appear to reduce Arc mRNA,²⁰² which is notable because it suggests a post-transcriptional mechanism.

The hippocampal GABA_Aergic interneuron population is notable for a high density of the α_5 -subunit subtype. Of known significance to memory function, the α_5 -GABA_A

receptor regulates the induction of LTP in response to a narrow range of frequencies in the theta range.²⁴² These observations have driven a series of anesthesia studies using α_5 -GABA_A-knockout mice. Etomidate blocks LTP measured in CA1 neurons in wild-type mice, but not in α_5 $-/-$ mutants, and behaviorally the α_5 $-/-$ mutants are resistant to the amnestic effects of etomidate, but not to its general anesthetic effects.²⁴³ Further, the effects of etomidate on LTP and memory behaviors are reversed by the addition of L-655,708, which selectively reduces the activity of α_5 -GABA_A receptors.²⁴⁴ The behavioral findings were later replicated with exposure to 1 MAC of isoflurane, with memory deficits observed in wild-type mice, but not in α_5 $-/-$ mutants or in wild types receiving L-655,708.^{245,246} The increase in α_5 -GABA_A receptors caused by etomidate and isoflurane does not return to baseline for at least 1 week.²⁴⁷

The α_5 -subunit subtype is not exclusive in its association with amnesia. α_4 -GABA_A receptors are concentrated in the dentate gyrus and dorsal thalamus, and α_4 knockouts are resistant to the amnestic effects of isoflurane but not to its general anesthetic effects.²⁴⁸ More equivocally, β_3 -GABA_A receptor knockout mice have been reported to be resistant to the amnestic effects of isoflurane,²⁴⁹ while in a study using β_3 -GABA_A receptor knock-in mutants, the β_3 subtype was not significantly associated with amnesia.²⁵⁰ Similar equivocal results have been found in studies of the α_1 subtype.^{251,252}

The effects of anesthetics on hippocampal theta oscillations *in vivo* have been studied. One study used a fear-conditioning paradigm to examine amnestic concentration of isoflurane, nitrous oxide, and halothane, and reported that suppression of hippocampal-dependent contextual conditioning was proportionate to slowing of theta peak frequency.²⁵³ Another study also demonstrated that isoflurane caused slowing of theta oscillations in the CA1 neuronal bundle without changing absolute power,²⁵⁴ while in contrast the nonimmobilizer F6, which causes amnesia without sedation or a loss of motor activity, caused a loss of theta oscillatory power without slowing. Scopolamine causes amnesia at doses associated with acceleration of theta oscillations, which might otherwise be expected to improve learning.²⁵⁵ However, a significant loss of absolute power is also observed. Together these findings suggest that individual anesthetics may cause amnesia through distinct forms of hippocampal theta disruption.

Human and Nonhuman Studies of Anesthetic Effects on Fear Memory Systems

THE AMYGDALA-DEPENDENT FEAR SYSTEM

The amygdala is a cluster of interconnected nuclei sitting immediately anterior to the hippocampus, with afferent and efferent projections to a wide distribution of cortical and subcortical structures. It is critical to fear learning and memory, and the systematic study of amygdala-dependent classical (Pavlovian) fear conditioning has produced much information regarding the mechanisms and circuitry.^{145,146} The basolateral nucleus of the amygdala (BLA) also modulates the encoding and consolidation of memory in the

hippocampus and elsewhere in response to emotion, arousal, and stress.²⁵⁶ The mechanism is dependent on noradrenergic projections terminating on both α and β receptors within the BLA,²⁵⁷ and can be triggered by systemic stress mediators—notably glucocorticoids²⁵⁸ and epinephrine.²⁵⁹ Amygdalo-hippocampal connectivity occurs via direct and indirect projections, and is dependent on theta oscillatory synchrony.^{181,260} The amygdala contains a number of rational targets for GABA-ergic anesthetic drugs.

STUDIES OF ANESTHETIC EFFECTS ON FEAR SYSTEMS IN NONHUMANS

As various forms of fear conditioning represent the dominant experimental method for studying memory in animal models, almost all studies of anesthetic effects on memory offer some insight into their effects on fear systems. Beyond the central conclusion from this sizeable body of literature that GABA-ergic anesthetics in general impair the acquisition of fear memory, a subset of studies are notable for their specific focus on fear circuitry and behavior.

There is substantial evidence that anesthetic effects on fear memory are mediated by GABA-ergic mechanisms in the amygdala. Selective injection of the BLA with midazolam blocks acquisition and stress enhancement of fear memory,²⁶¹ while lesions of the BLA block anterograde amnesia for an inhibitory avoidance task in rats receiving diazepam,²⁶² propofol,²⁶³ and sevoflurane.²⁶⁴ Further, injection of the BLA with the selective GABA_A antagonist bicuculline blocks the effect of propofol²⁰² and midazolam.²⁶⁵

There are certain conditions under which anesthetic drugs can enhance fear memory. When propofol or ketamine is given immediately following conditioning, retention is enhanced,^{266,267} which may represent a form of retrograde facilitation as described earlier, but retention is reduced when dexmedetomidine is similarly administered. In a rat model of PTSD, propofol and ketamine administered immediately after fear learning enhances long-term fear behavior, while dexmedetomidine has a neutral effect.²⁶⁷ Midazolam given during extinction training, which normally inhibits learned fear expression, blocks the effect of extinction,²⁶¹ and also blocks fear-reducing effects when given during preconditioning contextual learning.²⁶⁸ When sevoflurane is administered at a very low, nonsedating dose (0.11%), fear conditioning is enhanced.^{269,270}

STUDIES OF ANESTHETIC EFFECTS ON FEAR SYSTEMS IN HUMANS

Three functional neuroimaging studies have directly addressed anesthetic effects on emotional memory. A study using PET and path analysis (Fig. 9.5 lower panel)²⁷¹ demonstrated that the superior memorability that negatively arousing emotional items have over neutral items was unaffected by 0.1% and 0.2% sevoflurane, but lost at 0.25%, and was associated with decreased effective connectivity projecting from the right amygdala to hippocampus, and also from the right nucleus basalis to hippocampus. Further, sevoflurane modulated emotional perception such that items were rated more neutrally. An fMRI study demonstrated that sedative doses of propofol do not reduce activation of the amygdala in response to negatively arousing items, but activation of the

hippocampus is markedly attenuated (see Fig. 9.5 upper panel) and correlates with amnesia and loss of the superior memorability of emotional material.²¹⁸ This suggests that cortical and subcortical processes serving emotional interpretative functions and afferent amygdala activation are relatively spared by amnesic levels of propofol, while efferent processes underlying amygdala-dependent modulation of hippocampal plasticity are interrupted. An fMRI study of dexmedetomidine also reported no effect on activation of the amygdala in response to negatively arousing items,²¹⁶ but in contrast to propofol, emotional items retained superior memorability, and activation in the left amygdala and hippocampus were correlated with subsequent memory. Taken together, these studies imply that propofol may have a more targeted effect on the amygdalo-hippocampal modulatory axis. One potential explanation is that amygdalo-hippocampal connectivity is markedly interrupted by a loss of theta oscillatory synchrony caused by propofol, whereas α_{2A} antagonism at the LC causes only limited downstream attenuation of noradrenergic signaling in the BLA.

CLINICAL RELEVANCE

Amygdala reactivity is of importance in a number of fear-based psychopathologies, including anxiety, phobia, panic disorder, and PTSD.²⁷² The neurohumoral stress response to surgery and critical illness can cause complex changes in levels of catecholamines, glucocorticoids, and other potential promoters of plasticity in amygdala-dependent fear processes, occurring within a context that frequently contains emotional stressors. It is unclear under what specific conditions elevated (or depressed) levels of these mediators might contribute to long-term changes in neuropsychological functioning, although multiple studies in the critical care setting relate stress exposure, and exogenous administration of catecholamines, to negative long-term outcomes.²⁷³ In a broad surgical population, the incidence of PTSD-complex symptoms related to surgery was reported to be 16%⁵—similar to that observed in victims of violence. The effect—or lack of effect—of anesthetic drugs on plasticity processes in the perioperative or critical care setting thus has a theoretical potential to impact long-term psychological sequelae, both positively and negatively. However, there is presently insufficient human data to inform specific recommendations.

Anesthetic Effects on Implicit Memory Function

Implicit (nondeclarative) memory processes do not axiomatically require a conscious substrate, and are not dependent on classic hippocampal plasticity. Therefore the effect of anesthetic drugs on implicit memory functions may have important distinction from those on declarative memory, and accordingly a number of investigations have sought evidence for implicit processing in unconscious subjects. Many studies have used auditory adaptations of the *word stem completion task*, or the more rigorous *process dissociation procedure*, in which subjects are asked to exclude items that have been presented; declarative memory will drive avoidance of the target word, while implicit memory will lead to a familiarity response favoring the target word.

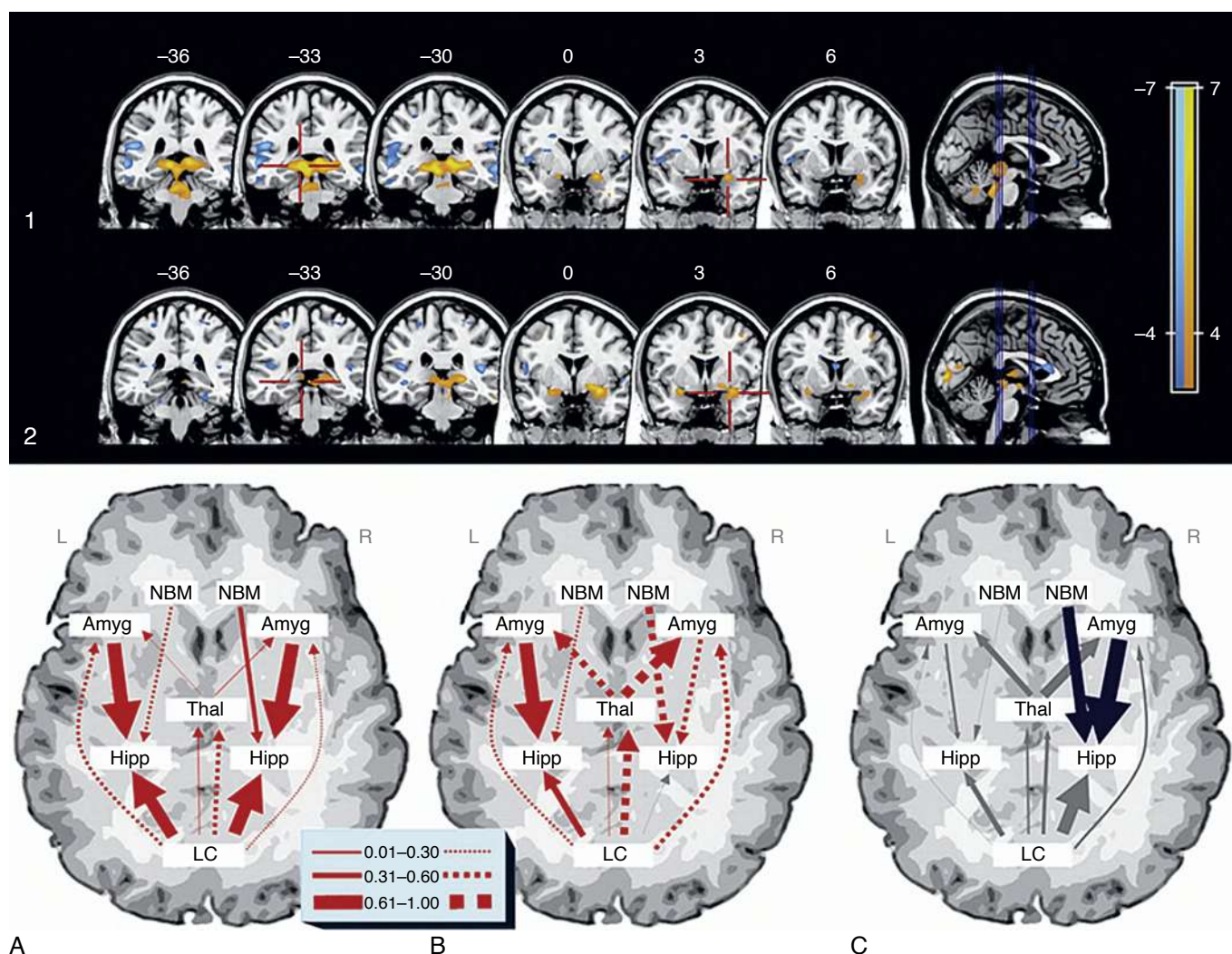


Fig. 9.5 Anesthetic effects on emotional memory systems. (*upper panel*) Functional magnetic resonance imaging scans showing the contrast in response to negatively arousing items relative to neutral items, with coronal slices through the amygdala (0, 3, 6) and hippocampus (−30, −33, −36). The control condition (*top row*) shows enhanced activation of both the amygdala and hippocampus in response to the added emotional information, while in subjects receiving propofol (*bottom row*) there is enhanced activation of the amygdala, but not of the hippocampus. (*lower panel*) Path diagrams of connectivity at rest, with positive influences of one region onto another shown as solid lines, negative influences shown as dotted lines, and width representing magnitude. In the control condition (*A*) there is a significant positive influence of the amygdala onto the hippocampus bilaterally. Administration of sevoflurane 0.25% (*B*) blocks emotional modulation behaviorally, and removes the positive influence of the right amygdala and nucleus basalis of Meynert on the hippocampus. Numerical differences in path weight (*C*) show these two paths to significantly contribute to the network model more in the control than sevoflurane state. Amyg, Amygdala; Hipp, hippocampus; LC, locus ceruleus; NBM, nucleus basalis of Meynert; Thal, thalamus. ([A] Modified from Pryor KO, Root JC, Mehta M, et al. Effect of propofol on the medial temporal lobe emotional memory system: a functional magnetic resonance imaging study in human subjects. *Br J Anaesth*. 2015;115[suppl 1]:i104–i113, Figure 3; [B] Modified from Alkire MT, Gruver R, Miller J, et al. Neuroimaging analysis of an anesthetic gas that blocks human emotional memory. *Proc Natl Acad Sci U S A*. 2008;105[5]:1722–1727, Figure 5.)

Two early studies in patients undergoing coronary artery²⁷⁴ and gynecologic²⁷⁵ procedures showed evidence of implicit learning using word stem completion tasks. Another early study demonstrating implicit recall in cardiac surgery patients correlated performance with preservation of the midlatency auditory evoked potential, and most strongly with the early cortical Pa and Na complexes.²⁷⁶ Later studies demonstrated implicit memory in both trauma patients²⁷⁷ and patients undergoing an emergency cesarean section²⁷⁸ using the process dissociation procedure, and further demonstrated that the degree of processing is related to the bispectral index. However, multiple more recent studies using similar procedures have either

failed to show significant implicit priming effects or have been equivocal.^{279–281} Synthesis of these disparate results into a cohesive conclusion is difficult. Although differences in patient populations and anesthetic regimens may be of relevance, methodological challenges may have driven both false positive and false negative results. Investigations evaluating implicit memory processes in pediatric patients under anesthesia have also offered conflicting results. One study reported an increased ability to distinguish a primed animal sound from white noise,²⁸² but other investigations found no evidence of priming.^{283–285}

KEY POINTS

- Sleep is a dynamic neuronal and behavioral state that can be characterized using specific electroencephalographic, electrophysiologic, and behavioral findings.
- Characteristics of sleep can be quantified using questionnaires, actigraphy, or respiratory polygraphy. However, polysomnography, including electroencephalogram, electrooculogram, submental electromyogram, and analysis of breathing, is required to describe the cortical characteristics and immediate physiological consequences.
- Activation of hypothalamic sleep-promoting pathways including the ventrolateral preoptic nucleus and the median preoptic nucleus produces the physiological switch from wakefulness to sleep.
- Sleep and anesthesia can look similar. Although small doses of anesthetics can induce sleep by activating sleep-promoting pathways, surgical anesthesia and immobilization cannot be induced by sleep-promoting pathways.
- Control of breathing is altered during sleep and anesthesia, typically leading to a decreased drive to upper airway dilator and respiratory pump muscles.
- Persistent respiratory depressant effects of anesthetics and neuromuscular blocking drugs increase the risk of postoperative respiratory complications, particularly in patients with obstructive sleep apnea.
- Anesthesia and surgery, as well as treatment in the intensive care unit and opioids, affect sleep duration and sleep architecture, which may lead to impaired outcome.

Probably the earliest mention of “overpowering sleep” as a metaphor describing what possibly characterizes anesthesia can be found in Genesis 2:21: “And the Eternal God caused an overpowering sleep to fall upon the man and he slept. He took one of his ribs and shut in flesh instead thereof.” Although this “overpowering sleep” allegedly occurred by divine intervention and did not represent the drug-induced unconsciousness and immobility that characterizes anesthesia, the basic narrative is an example of antiquarian history that implicates the concept that deep sleep may be the only viable condition that allows a surgical procedure to be completed successfully.¹

In the 21st century, an increasing body of knowledge has revealed the common ground between sleep and anesthesia, as well as their fundamental differences in clinical picture and underlying mechanisms.

Sleep is required for survival. Rats deprived of sleep will die within 2 to 3 weeks, a time frame similar to death due to starvation.² In humans, sleepiness can be deadly. Approximately 100,000 motor vehicle accidents each year result from drivers who are “asleep at the wheel.” In a survey of drivers in New York State, approximately 25% reported they had fallen asleep at the wheel at some time.³ In addition, sleep deprivation impairs the psychomotor performance of night shift workers, specifically surgeons and residents. In patients, perioperative sleep deprivation occurs frequently, particularly in critically ill patients, and there may be consequences on patients’ outcomes. Among sleep disorders, obstructive sleep apnea (OSA) has probably the most meaningful consequences for perioperative treatment.⁴

Therefore, high-quality physiologic sleep is a key domain of public health that needs to be considered in perioperative medicine from both a provider and a patient perspective.

Definition of Sleep

Sleep is the natural periodic suspension of consciousness during which the powers of the body are restored. A behavioral definition of sleep includes a species-specific posture and behavioral quiescence, as well as elevated arousal thresholds. However, sleep is much more than an absence of activity. The brain is highly active during sleep, especially during rapid eye movement (REM) sleep, during which atonia, phasic muscle movements (driven by different activation levels in distinct areas of the brain), and vivid dreaming occur. Sleep is not a simple electrophysiological phenomenon. At different stages of sleep, brain activities can be as distinct from each other as they are from wakefulness.⁵ In humans and other vertebrates (see “Evolution”), sleep has two major stages: REM sleep and non-REM (NREM) sleep, with each of these further divided into substates. Most of sleep time is spent in NREM sleep, which is characterized by relatively low electroencephalographic (EEG) frequencies with higher amplitudes compared with awake EEG frequencies (Fig. 10.1). In the transition from wake to NREM sleep, the fast activity of waking disappears (α - θ transition), and then in the deeper stages of NREM large, slow waves (i.e., δ waves) emerge. Thus deep NREM sleep is also referred to as *slow-wave sleep*. NREM sleep is associated with waxing and waning muscle tone, as well as decreased body

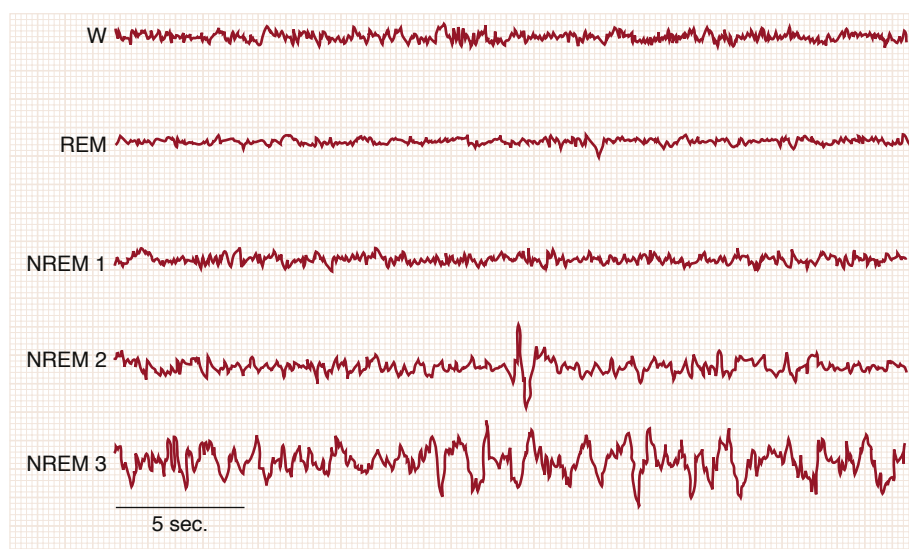


Fig. 10.1 Representative electroencephalogram activity seen during different behavioral states. Electroencephalogram recordings of one patient during wakefulness (eyes closed; W), rapid eye movement sleep (REM), and non-REM (NREM) sleep stages 1 to 3.

temperature and heart rate. In contrast, REM sleep is characterized by muscle atonia, loss of slow waves in the EEG, and bursts of REMs that give this state its name.⁶ Other prominent features of REM include marked irregularities of respiration and heart rate, as well as penile and clitoral erection. REM sleep is associated with the high likelihood of vivid dreaming. A distinct property of REM sleep is the operation of a system that suppresses motor activity, which is responsible for baseline atonia and suppression of motor commands that would otherwise result in acting out of dreams, a phenomenon known as *REM sleep behavior disorder*. This REM atonia system is not constant, as it periodically allows breakthrough muscle activity that comprises REM sleep and twitching of the extremities. REM sleep has therefore been subdivided into *tonic REM sleep* (i.e., a period with muscle atonia and without eye movements) and *phasic REM sleep*, which is interrupted by short phasic events of the eye and other movements.⁷ The pattern and amounts of NREM sleep, REM sleep, and wakefulness throughout the night are referred to as *sleep architecture*, and there are many physiologic and pathophysiologic processes that can affect sleep architecture. For example, many antidepressant medications, benzodiazepines, and opioids selectively suppress REM sleep. A disease example is narcolepsy, in which individuals often transition rapidly from wake to REM, whereas normally REM is nearly always entered from a prolonged episode of NREM state.

Physiology

EVOLUTION

It is unknown why we sleep; however, one can imagine that evolutionary pressure would favor systems that are optimized for nature's rhythms.^{8,9} In a rhythmic world, organisms must adapt to alternating changes in the environment, such as daily and seasonal rhythms in light intensity, ambient temperature, and humidity.

CIRCADIAN RHYTHMS

Species-dependent temporal behavior can be adapted depending on the environmental conditions: behaviorally, anatomically, and physiologically. Circadian rhythms affect almost every aspect of the body's function, including activity and rest patterns, cognitive function (e.g., learning and memory), cardiovascular and endocrine physiology (e.g., heart rate, metabolism, and hormone secretion), and gene expression (15% of the genes in the human body show daily rhythms).

Virtually all species show circadian rhythms that regulate periodic changes in behavioral and physiologic parameters within a period of approximately 24 hours.¹⁰ These rhythms can be characterized as a synchronization of activity to the external light-dark cycle in the majority of living creatures. Bacteria, plants, animals, and humans exhibit such a behavior that helps them stay in tune with the environmental light-dark cycles.¹¹ Our sleep timing preference, or chronotype, is a manifestation of our internal biological clock. Variation in chronotype has been linked to sleep disorders, cognitive and physical performance, and chronic disease.¹²

Clock genes generate endogenous clock pulses in most (if not all) cells of the body, synchronized by regulatory pathways across the body to a superior rhythm generator (a so-called master clock). Desynchronization of these rhythms seems to be involved in the pathogenesis of metabolic, psychiatric, and other disorders.¹¹ The superior rhythm generator in humans is located in the suprachiasmatic nucleus that receives external inputs about light and darkness from the retinal cells and synchronizes inputs of melatonin levels. While this master clock synchronizes the behavioral and biological rhythms of the human body to the changing demands of the environment during the solar day, sleep itself entrains circadian rhythms,^{11,13} and the temporal organization of circadian rhythms and nocturnal sleep must be preserved to accomplish a restful and refreshing experience that fulfills the sleep needs produced by the

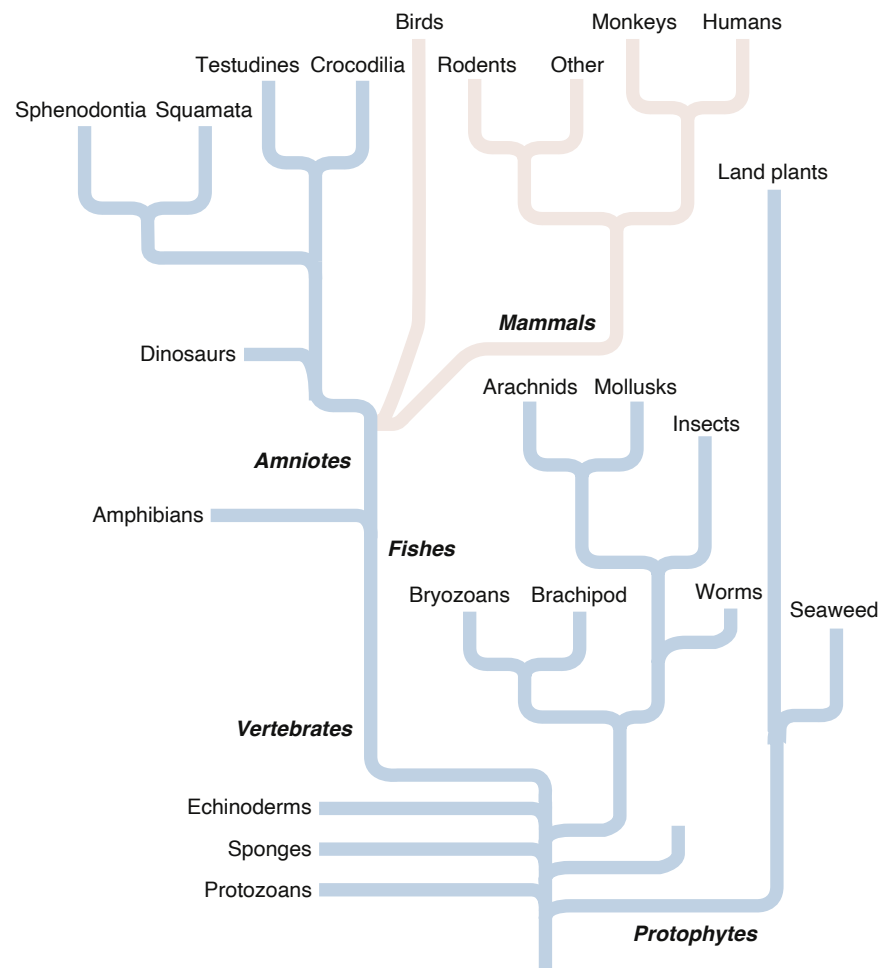


Fig. 10.2 Evolutionary tree. Circadian rhythms are common among all living beings. However, only mammals and birds (red) have integrated sleep with different stages. Fish, reptiles, insects, and plants (blue) manifest circadian rhythms as periods of activity and rest.

second major regulatory factor, sleep homeostasis. During waking, “sleep pressure” continuously increases, leading to an increased likelihood for the transition from wakefulness to sleep.¹⁴ Increasing homeostatic sleep pressure can be partly offset by a sufficient circadian waking stimulus during the circadian wake phase, but the circadian wake stimulus disappears in the presence of an overwhelming sleep pressure.¹⁵ In this situation, sleep sufficient in quantity and quality is needed to reestablish proper functioning.

Although circadian and homeostatic regulation of sleep can be observed in almost all living organisms,¹⁶ differentiation of sleep into NREM and REM sleep (“integrated sleep”) probably has developed during the last 300 to 350 million years because it appears only in birds and the majority of terrestrial mammals (Fig. 10.2).^{17,18}

Sleep Stages and Sleep Cycles

Sleep architecture—a structured temporal order of different sleep stages—is an important determinant of sleep quality. If tired individuals rest in their individual sleeping postures, the increasing level of drowsiness is paralleled by continuous EEG slowing from that seen during attention and cortical activation (16–30 Hz, “ β activity”) to slower frequencies, predominantly between (7.5 and 11 Hz, “ α activity,” typically observed in subjects with reduced attention and closed eyes).

During this state of vigilance, the individuals do not sleep, and a full level of cognitive function can be established easily even without a strong sensory stimulus. The transition from α state to sleep is paralleled by a further decrease in α and β activity and an increase of EEG activity in the θ band (4.5–7.5 Hz; Fig. 10.3).

This transition from α to θ activity is widely accepted as the EEG correlate of sleep onset.¹⁹ At this transition from wakefulness to NREM sleep stage 1, heart rate and heat production decrease, resulting in a slight decrease in body temperature. Furthermore, respiration is regular and deep. As sleep deepens, EEG activity shows low amplitudes and θ activity with intermittent sleep spindles and K-complexes. The latter represents an activation of brainstem and subcortical brain areas during sleep^{20,21} for selective processing of unexpected sensory inputs (e.g., sounds) that might require complete arousal and restoration of consciousness to address a potential threat.²² In accordance with that hypothesis, in comatose patients with brain injury, the presence of K-complexes in response to an acoustic stimulus during coma seems to be a marker of a better outcome compared with comatose patients with no evoked K-complexes.²³ With increasing depth of sleep, during NREM sleep, electrical activity of the cortex measured by EEG is dominated by slower frequencies and high amplitudes, also referred to as *slow-wave sleep*.



Fig. 10.3 Electroencephalogram recording during sleep onset (arrow). The two top rows show bipolar electroencephalogram (EEG) signals from left (C3-A1) and right (C2-A1) frontal leads. Rows 3 to 5 show the relative amount (EEG power) of β , α , and θ activity calculated using fast Fourier transformation.

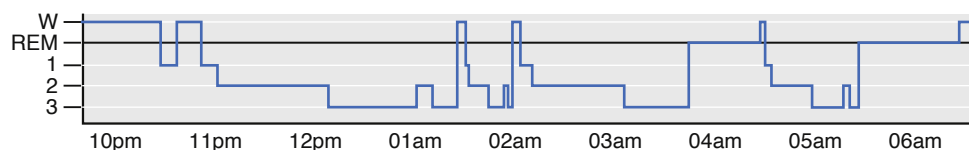


Fig. 10.4 Hypnogram of a night of physiologic sleep. During one night (time on x axis) of sleep, humans transition repetitively from one sleep stage to another with short episodes of occasional wakefulness (indicated on y axis). REM, Rapid eye movement sleep; W, awake; 1, non-REM sleep stage 1; 2, non-REM sleep stage 2; 3, non-REM sleep stage 3 (slow wave sleep).

REM sleep, or paradoxical sleep, is associated with an alteration of homeostatic regulation, such as increased heart rate variability, irregular respiration, and impaired body temperature control. Brain metabolism increases while EEG recordings show similarities to wakefulness with low-voltage and mixed-frequency power spectra. Prominent θ waves generated by the hippocampus occur. These waves are not apparent in the scalp EEG recording from humans, but in rodents, in which the hippocampus is larger and closer to the brain surface, the EEG is dominated by θ during REM. This sleep state is associated with a decreased tone in skeletal muscles (except for extraocular muscles that control eye movement). Dreaming is a typical experience during REM sleep,²⁴ but can also happen during NREM sleep.²⁵ During physiologic sleep, an individual typically switches back and forth across sleep stages, interrupted by occasional arousals from sleep (Fig. 10.4).

NEUROANATOMY OF SLEEP

Sleep-Promoting and Arousal Pathways

Several neuronal pathways have evolved to maintain cortical activation and behavioral arousal during normal waking, and others have evolved to promote and maintain sleep

(Fig. 10.5). It is the balance between these two systems that determines whether one is awake or asleep.

Ascending Arousal System. The ascending arousal system (AAS) is the main wakefulness-mediating network in the brain. Main pathways within the network receive cholinergic, monoaminergic, dopaminergic, and glutamatergic inputs. Cholinergic inputs originate from pedunclopontine and laterodorsal tegmental nuclei, innervate the lateral hypothalamus, prefrontal cortex, basal forebrain, and thalamic relay nuclei (i.e., medial and lateral geniculate nuclei; mediodorsal nucleus; pulvinar; anterior, ventral, and lateral thalamic cell groups).^{26,27} Glutamatergic neurons that provide inputs to the AAS are mainly located within a small area just ventral of the locus ceruleus (LC), the preceruleus area, as well as in the parabrachial nuclei that mainly project to the basal forebrain and the lateral hypothalamus.^{28,29} Monoaminergic inputs mainly come from noradrenergic neurons within the LC, histaminergic neurons of the tuberomammillary nuclei (TMN), serotonergic neurons of the median and dorsal raphe nuclei,^{30,31} as well as dopaminergic neurons adjacent to dorsal raphe nuclei.^{24,32} Beside projections to the basal forebrain, multiple inputs are sent to the thalamus, mainly the interlaminar

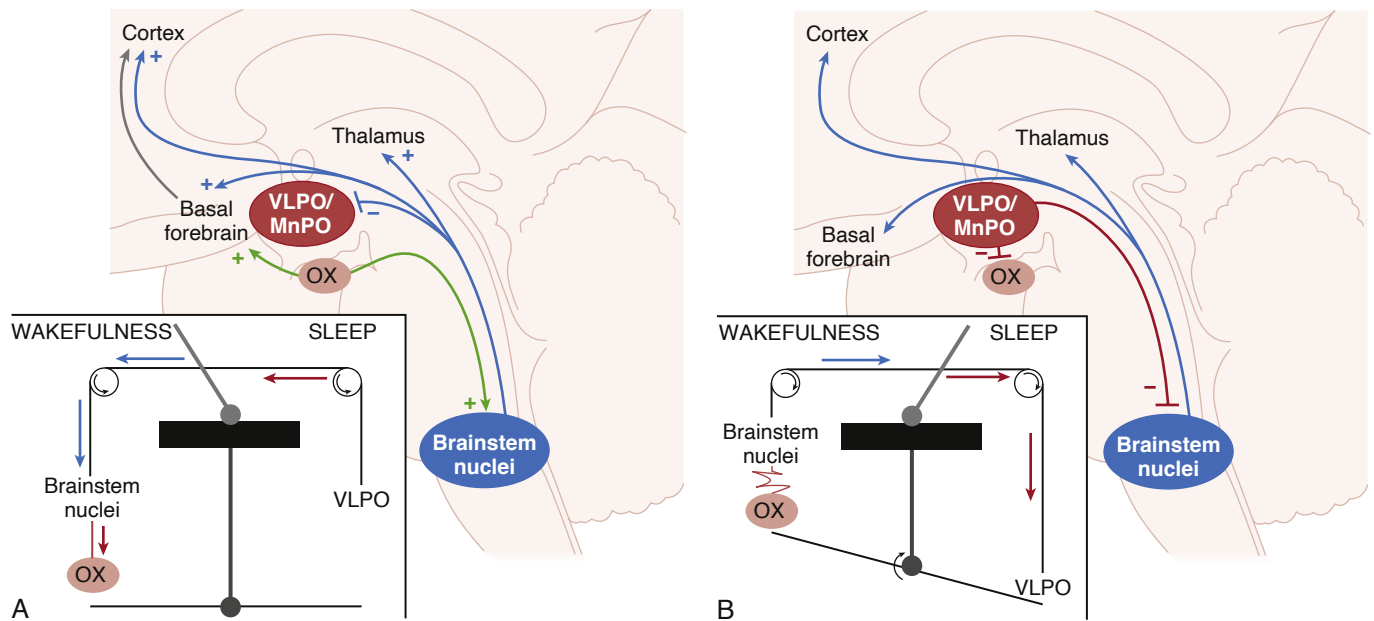


Fig. 10.5 Flip-Flop switch ensuring rapid transitions between wakefulness and sleep. During wakefulness (A), brainstem nuclei of the ascending arousal system (AAS) directly and indirectly provide excitatory input to the thalamus, basal forebrain, and the cerebral cortex, while they inhibit the ventrolateral (VLPO) and median (MnPO) preoptic nuclei (switch flips to wakefulness). The excitatory, wake-promoting stimulus is enforced by additional excitatory input of orexinergic neurons (OX) to the BF and the AAS. During sleep (B), neurons of VLPO and MnPO inhibit the brainstem and orexinergic neurons of the AAS (switch flips to sleep). (Modified from Saper CB, Scammell TE, Lu J. Hypothalamic regulation of sleep and circadian rhythms. *Nature*. 2005;437:1257–1263; and Saper CB, Fuller PM, Pedersen NP, Lu J, Scammell TE. Sleep state switching. *Neuron*. 2010;68:1023–1042.)

and reticular nuclei, as well as the amygdala and cerebral cortex. Furthermore, noradrenergic neurons of the LC project mainly to the posterior part of the lateral hypothalamus. In return, the latter area of the lateral hypothalamus projects to the LC and to the TMN.

Some neurons of the posterior hypothalamus produce orexin A and B, also known as hypocretin (HCRT) 1 and 2. These neurons project to the basal forebrain and the amygdala and cerebral cortex and other important arousal areas, and they are essential for stabilizing the waking state.³² Orexin deficiency causes narcolepsy (severe daytime sleepiness) with cataplexy (abrupt loss of muscle tone),³³ a disorder showing some characteristics of REM sleep (i.e., REM sleep atonia during wakefulness [discussed later]).

Orexin System. Orexin and its cognate receptors (OX₁R and OX₂R) play a key role in sleep/wake regulation. This neurotransmitter system has a crucial involvement in the pathogenesis of narcolepsy, a disorder associated with chronic excessive daytime sleepiness, sleep attacks, and rapid transitions between vigilance states and sleep in humans.³⁴ Studies in mice recapitulating the narcolepsy phenotype by targeted mutagenesis of the HCRT gene encoding the prepropeptide precursor of orexin neuropeptides,³⁵ or the less pervasive phenotype of OX₂R knockout mice,³⁶ not only demonstrated the role of orexin signaling in sleep/wake regulation but also demonstrated biological proof of concept for the development of orexin receptor antagonists for the treatment of insomnia. In contrast, the activation of orexin pathways, either by exogenous administration of orexin receptor agonists or by optogenetic activation of orexinergic neurons in the lateral hypothalamus, increases arousal.^{37,38} Orexinergic neurons are counteracted by inhibitory γ -aminobutyric acid (GABA)-ergic

neurons originating from the ventrolateral preoptic nucleus (VLPO). OX₂R is expressed exclusively in the histaminergic TMN, whereas OX₁R is expressed preferentially in the noradrenergic LC, and both receptors are present in cholinergic pedunculopontine and laterodorsal tegmental nuclei.^{38,39} Orexinergic tone at these hypothalamic and brainstem nuclei contributes to cortical arousal.⁴⁰ TMN, the predominant source of histamine in the brain responsible for cortical arousal, receives a prominent projection from orexinergic neurons^{40,41} mediated predominantly by OX₂R. Orexin receptor antagonists promote sleep mainly via inhibition of the OX₂R receptor.⁴²⁻⁴⁵

Non-Rapid Eye Movement Sleep-Promoting Pathways.

About 100 years ago during the epidemic of encephalitis lethargica, it was observed that lesions in the preoptic region around the rostral end of the third ventricle were associated with profound insomnia,⁴⁶ which was confirmed in neuroanatomic experiments (lesion studies) in rats and cats.^{47,48} Neurons in the preoptic area are active during sleep.⁴⁹⁻⁵¹ Two key nuclei within this area are the VLPO as well as the median preoptic nucleus. VLPO neurons fire at a higher frequency during sleep compared with wakefulness.⁵² Anatomically, the VLPO consists of a dense core of sleep-active, galanin-positive neurons innervating the TMN (which is part of the AAS) while being surrounded dorsally and medially by a more diffuse population of sleep-active, galanin-positive neurons that project to the dorsal raphe and LC.⁵³ Physiologically, the VLPO neurons constitute a sleep-promoting pathway that inhibits many components of the arousal system during sleep. Likewise, parts of the arousal system such as the laterodorsal tegmental nucleus and the pedunculopontine tegmental nucleus, as well as the LC, the parabrachial nucleus, the dorsal raphe nucleus, the PC,

the ventral periaqueductal gray, and the TMN, are capable of inhibiting the VLPO. The mutually inhibitory relationship of the arousal and sleep-promoting pathways produces the conditions of a flip-flop switch, which can generate rapid and complete transitions between waking and sleeping states (see Fig. 10.5).^{53,54} It also makes simultaneous activation of arousal and sleep circuits highly improbable.

Because animal experiments showed that even large lesions of the VLPO substantially reduce but do not completely erase sleep, other brain areas beside this nucleus are likely involved in promoting sleep.⁵⁵ Several basal forebrain areas,^{49,51} as well as some GABA-ergic interneurons across the cerebral cortex,⁵⁶ have been implicated to function as sleep-active neurons. Nevertheless, the role of these brain regions in the promotion or regulation of sleep remains unclear.

Rapid Eye Movement Sleep-Promoting Pathways and Non-Rapid Eye Movement–Rapid Eye Movement Transition. During normal sleep that presents at onset as a sharp α - θ EEG activity transition, a similar sharp transition from NREM to REM sleep can be observed. Two groups of mutually inhibitory neurons located in the pons are involved in mediating the switch between NREM and REM sleep.²⁸ The first group consists of REM active inhibitory neurons of the sublaterodorsal nucleus and the PC region.^{29,54,57} These neurons inhibit but are also inhibited by a second group of neurons that are located in ventrolateral periaqueductal gray and adjacent lateral pontine tegmentum. This mutually inhibitory relationship produces a REM-NREM flip-flop switch, promoting rapid and complete transitions between sleep states.⁵⁴

Within the sublaterodorsal nucleus and PC, glutamatergic neurons are mixed within the population of REM-on GABA-ergic neurons. The glutamatergic sublaterodorsal nucleus neurons project to the spinal cord and are important for REM sleep atonia; the glutamatergic PC neurons activate forebrain pathways driving EEG desynchronization and hippocampal θ rhythms—characteristic EEG signs of REM-sleep.⁵⁸

How to Assess Sleep

The complexity of signs and symptoms of sleep suggests that different instruments need to be applied to capture all the important elements of sleep.

QUESTIONNAIRES

Several questionnaires are used for the assessment of sleep duration, sleep quality, and the related physiologic and pathophysiologic consequences. Quality of sleep is a frequent target of generic health surveys for measuring patient-reported outcomes. Other questionnaires quantify the consequences of sleep deprivation, specific sleep disorders, or both (Table 10.1). The most commonly used instrument in sleep medicine is probably the Epworth Sleepiness Scale, a short questionnaire to assess symptoms of daytime sleepiness, expressed as intolerance to monotony. The questionnaire asks the subject to rate the probability of falling asleep on a scale of increasing probability from 0 to 3 for eight different situations, such as watching TV, reading, lying down in the afternoon, and so on.⁵⁹ Despite the fact that the Epworth Sleepiness Scale

TABLE 10.1 Sleep Questionnaires: Questionnaires Commonly Used for Clinical and Research Purposes and Their Focus

Questionnaire*	Focus	References
<i>Pittsburgh Sleep Quality Index (PSQI)</i>	Sleep and sleep disorders	60,61
Sleep Quality Scale (SQS)	Sleep quality	66
Functional Outcome of Sleep Questionnaire (FOSQ)	Impact of daytime sleepiness on daily living	12,67
Pediatric Sleep Questionnaire	Sleep and SDB in children	68
Child Sleep Habits Questionnaire (CSHQ)	Sleep	69
<i>Epworth Sleepiness Scale (ESS)</i>	Daytime sleepiness	59,62,63
<i>Sleep Diaries/Sleep Logs</i>	Sleeping times, sleep duration	70,71
<i>Morningness-Eveningness Questionnaire</i>	Sleeping times, sleep duration, circadian rhythm	72
Loughborough Occupational Impact of Sleep Scale (LOISS)	Sleep quality	73
Insomnia Sleep Questionnaire (ISQ)	Insomnia	74
Berlin Questionnaire	SDB in surgical patients	66
<i>STOP/STOP-Bang Questionnaire</i>	Obstructive sleep apnea	75,76
Brief Insomnia Questionnaire (BIQ),	Insomnia	77
<i>International Restless Legs Syndrome Study Group Rating Scale (IRLS)</i>	Restless leg syndrome	78
<i>Clinical Global Impression (CGI) Scale</i>	Restless leg syndrome	67
<i>Richard–Campbell Sleep Questionnaire</i>	Sleep assessment in the ICU	79

*Recommended questionnaires are italicized.

ICU, Intensive care unit; SDB, sleep-disordered breathing; STOP, snoring, tiredness, observed apneas, and high blood pressure.

is reliable for detection of daytime sleepiness,^{60,61} the instrument does not determine the mechanism of daytime sleepiness.^{62,63}

Other clinically useful questionnaires that focus on the detection of signs and symptoms of sleep disorders are available,⁵⁹⁻⁷⁹ which is important because the prevalence of sleep pathologies typically varies in a wide range across populations (see [Table 10.1](#)).

An effective process for clinical assessment of sleep-related diseases is a stepwise assessment using a screening tool first (e.g., Epworth Sleepiness Scale), followed by a more specific questionnaire that helps identify individual mechanisms or consequences of sleepiness.

A special form of sleep questionnaires comes in the form of sleep diaries and “morningness-eveningness” questionnaires that evaluate daily sleep habits, including sleep time, sleep duration, number of nocturnal awakenings, and subjective sleep quality.⁸⁰

Although questionnaires allow a quick and easy screening for daytime symptoms of sleep disturbances, they do not quantify sleep architecture. Subjective methods of assessing sleep are influenced by the spectrum of disease within a tested group, actual clinical change over time, the testing conditions, and recall bias.⁸⁰ Clearly, a questionnaire cannot replace a medical history or objective assessment for sleep disorders. However, sleep questionnaires are important tools for measuring health improvement or decline, predicting medical expenses, assessing treatment effects, or comparing disease burden across populations.

ACTIGRAPHY

Actigraphy is used to study sleep-wake patterns by detecting motion of the wrist with linear accelerometers in single or multiple axes. Based on movement-derived data, predictions of the time spent during sleep and wakefulness can be made, and even assumptions on sleep staging are made. Actigraphy can be used conveniently in a patient’s home for several nights, weeks, or longer,⁸¹

but its validity for detecting transitions from wakefulness to sleep has been debated, especially in patients with high levels of sleep fragmentation.^{81,82} In the clinical setting, actigraphy has been used to evaluate sleep patterns in patients with insomnia, for diagnosis of circadian rhythm disorders (including shift work), and in evaluating sleep in individuals who are less likely to tolerate polysomnography (PSG), such as infants and demented elderly. However, its accuracy is limited for patients with reduced mobility, such as nursing home or intensive care unit (ICU) patients. Actigraphy allows for convenient follow-up measurements to evaluate the effects of treatments designed to improve sleep architecture and circadian rhythm disorders.⁸³ In more recent years, an increasing number of wearable devices for the detection and analysis of activity and rest have been marketed. Although some have claimed to detect sleep accurately, as well as some sleep stages (mainly REM sleep),⁸⁴ none of the current models are yet approved or validated for clinical use.

RESPIRATORY POLYGRAPHY

Home respiratory polygraphy (RP) can be a cost-effective alternative to PSG for the diagnosis of sleep apnea-hypopnea syndrome. RP usually calculates airflow through nasal pressure changes taken via nasal cannulae. Thoracic and abdominal movements are measured by piezoelectric bands, which also measure body position and blood oxygen saturation by pulse oximetry. Using these parameters, RP can identify apnea and hypopnea, and categorize the pathophysiology of apnea into obstructive versus central. Furthermore, similar to actigraphy, RP provides some analysis of sleep duration and time spent awake, using its body position and light sensors. However, the quality of the latter can be more precisely evaluated by a combination of sleep diary and actigraphy. Nevertheless, EEG is required to decide whether a subject was awake or sleeping ([Table 10.2](#)).

TABLE 10.2 Pros and Cons of Different Methods of Sleep Evaluation

Method	Pros	Cons	Recommendation
Sleep questionnaire	Low cost Good compliance Easy to use	Recall bias Limited reliability in some populations Limited validity	Should be combined with interview
Sleep diary	Less recall bias Easy to administer Low cost Documentation of daily variability	Lower compliance compared with other questionnaires Biased by daily mood and expectations on sleep	Should be combined with other measurement of sleep duration (e.g., actigraphy)
Actigraphy	Provides objective information about daily variability and sleep duration Provides information on sleep habits at home Not influenced by patient expectations, recall bias, or memory impairments Lower cost than laboratory polysomnography	Limited assessment of sleep stages and sleep onset latency Higher cost than questionnaires	Should be combined with other measurement of sleep duration (e.g., sleep diary)

Continued

Method	Pros	Cons	Recommendation
Polygraphy	Lower cost than laboratory polysomnography Objective assessment of respiratory events Provide information on sleep habits at home	Limited assessment of sleep Higher cost than questionnaires Limited assessment of sleep disorders other than sleep-disordered breathing	Should be accompanied by questionnaires and clinical interview
Laboratory polysomnography	Objective assessment of sleep, sleep stages, and sleep disorders	High cost First-night effect Limited capacity No information about sleep habits at home	Should be last step in sleep evaluation with previous questionnaires and ambulatory screening (e.g., polygraphy)
Out-of-center polysomnography	Objective assessment of sleep, sleep stages, and sleep disorders Less first-night effect Lower cost than laboratory polysomnography Can provide information about sleep habits at home	Limited observation may lead to decreased recording quality	Good patient education mandatory If possible, surveillance recording preferred

Adapted in part from Martin JL, Hakim AD. Wrist actigraphy. *Chest*. 2011;139(6):1514–1527.

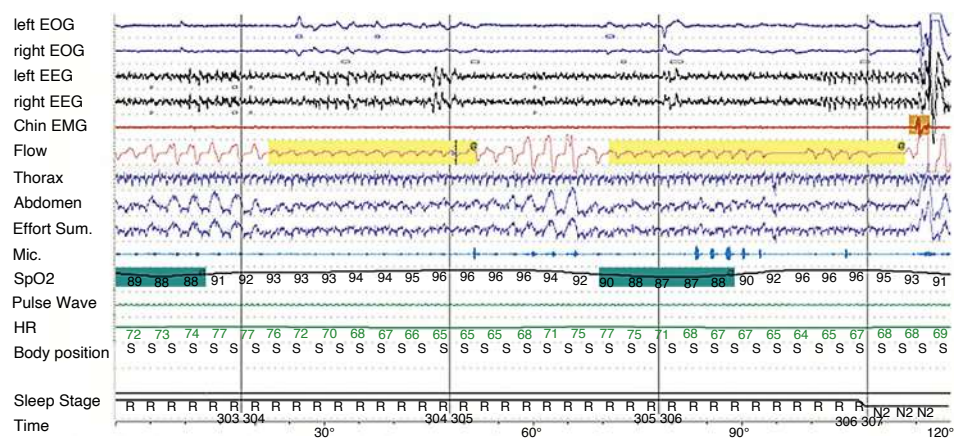


Fig. 10.6 Polysomnographic recording. Left and right electrooculogram (EOG), two electroencephalogram (EEG) channels, and mental electromyogram (Chin EMG) allow scoring of sleep stage (REM sleep [R] with typical eye movements in the electrooculogram channels). Additional channels (e.g., respiratory flow [Flow], respiratory effort of thorax [Thorax] and abdomen [Abdomen], microphone [Mic.] oxygen saturation [SpO₂]) allow diagnosis of sleep disordered breathing. In this patient, apneas (yellow boxes) led to oxygen desaturation (blue boxes) and finally to arousal (far right, brown box).

POLYSOMNOGRAPHY

PSG is the only method that can precisely determine the actual sleep stage and is a reference diagnostic tool (i.e., gold standard) that is required for the diagnosis of several sleep disorders.⁸⁵ PSG measurements include EEG, electrooculography for the measurement of eye movements, and at least electromyographic measurement of muscle activity of the chin.⁸⁶ These three measurements are typically supplemented by other channels for the detection of sleep-disordered breathing (SDB), such as a nasal sensor to detect apneas and hypopneas, oximetry, inductance plethysmography for respiratory effort of the chest and abdomen, and a body position sensor and leg electromyogram⁸⁷ to identify periodic limb movement syndrome or REM sleep behavior disorder, respectively.⁸⁸ A typical PSG recording is shown in Fig. 10.6.

Sleep Laboratory Testing

For several decades, PSG was performed only within a sleep laboratory. Sleep laboratory PSG is attended by a trained technician and is analyzed by a clinician trained in sleep medicine according to the published guidelines.^{87,89-91} However, the high cost and requirement of an overnight stay can present socioeconomic challenges that affect the accessibility of this method. In-center PSG provides only a momentary view of the patient's sleep, and the measurements themselves can impair patients' sleep (the so-called first-night effect) and does not provide information about sleep at home. Multiple repeated measurements to tailor treatment over time are challenging or even impossible from a logistic and healthcare economics perspective. Therefore, an increasing number of out-of-center devices

TABLE 10.3 Polysomnographic Characteristic of the Different Behavioral States

	EEG and EOG Characteristics	AASM
Wakefulness	α rhythm (8-13 Hz) present in more than 50% of an epoch	W
Non-REM sleep stage 1	Lower amplitudes and activity in range of 4-7 Hz (vertex sharp waves,* slow eye movements*)	N1
Non-REM sleep stage 2	EEG of sleep stage 1 with additional a sleep spindles and K-complexes; slow waves not fulfilling the criteria for sleep stage 3	N2
Non-REM slow wave sleep	20%-50% of slow wave activity (0.5-2 Hz)	N3
REM sleep	EEG of low amplitude and mixed frequencies, low chin electromyogram activity, rapid eye movements	R

*Not necessary but may be present.

AASM, American Academy of Sleep Medicine criteria (see reference 87); EEG, electroencephalography; EOG, electrooculogram; N1, non-REM sleep stage 1; N2, sleep stage 2; N3, sleep stage 3; R, REM-sleep; W, wakefulness.

are now available, offering the possibility of recording PSG in almost every environment, such as at the patient's home, within the hospital or nursing home room, and even within the recovery room or ICU.

Out-of-Center Testing

Portable out-of-center PSG sleep monitors are a reasonable, reliable, and effective method for diagnosing OSA⁹² compared with PSG recording performed within a sleep laboratory, when applied in preselected population at risk of moderate-to-severe OSA.⁹²⁻⁹⁴ The Portable Monitoring Task Force of the American Academy of Sleep Medicine (AASM) supports the use of these devices in subjects for clinical and research purposes and indicates a comparable quality of PSG data.⁹²

Scoring of Sleep and Sleep Disordered Breathing

More than 40 years ago, Rechtschaffen and Kales⁸⁶ standardized the method of scoring PSG recordings—the R&K criteria. Since then, major innovations in technology have transformed the science and clinical practice of sleep medicine. The latest criteria of the AASM^{87,95} capture the full potential of innovations resulting from computerization of data, including automation of sleep scoring, recognition of disorders that occur during sleep, and integration with other procedures, such as positive airway pressure titration. Although the R&K criteria are still sufficient for clinical and research purposes, they are less commonly used in sleep centers around the world.

According to these criteria, the arousal states are categorized based on the EEG activity recorded from three different scalp locations and divided into wakefulness (W), REM sleep (R), and three NREM sleep stages (Table 10.3 and see Fig. 10.1).

Respiratory events during sleep are defined as follows:

1. *Apnea* with a decrease in respiratory flow of 90% or more from baseline for at least 10 seconds, whereas a minimum of 90% of the event duration has to meet the criteria of respiratory flow reduction
2. *Hypopneas*, in which the signal of respiratory flow measurement decreases at least 30%, whereas oxygen saturation (SpO_2) decreases by 4% or more compared with pre-event baseline for at least 90% of the duration (minimum of 10 seconds). Hypopneas can also be defined as a decrease in respiratory flow measurement of at least 40% with a decrease in SpO_2 of only 3%.
3. *Respiratory event-related arousal* is defined as a series of breaths not meeting the criteria for apnea or hypopnea lasting at least 10 seconds characterized by increasing respiratory effort or flattening of the nasal pressure waveform leading to an arousal from sleep.
4. *Hypoventilations* are defined as an increase of 10% or more in arterial carbon dioxide tension (PaCO_2).

Although the AASM 2007 criteria⁸⁷ are commonly used in many sleep centers for clinical and research purposes all over the world, questions about whether the new criteria should be used in children⁹⁶ and the scoring of some respiratory events are still unanswered.

Sleep and Breathing

RESPIRATORY REGULATION DURING SLEEP

Sleep puts breathing at risk. The upper airway dilator muscle activity is decreased during sleep compared with wakefulness, particularly at sleep onset and during REM sleep. Furthermore, the ventilatory response to hypoxia can be impaired, such that critical hypoxia levels can occur during sleep that can be offset only by arousal from sleep.

The major determinant of minute ventilation during wakefulness and sleep is PaCO_2 . In contrast to wakefulness, where PaCO_2 is maintained close to 40 mm Hg, the chemo sensitivity to CO_2 decreases during sleep, as does the chemo sensitivity for oxygen. This leads to PaCO_2 values of 45 mm Hg commonly occurring during stable sleep, while ventilatory demand is decreased and the arousal threshold varies with sleep stages.

Accordingly, the changes in respiratory muscle activity, ventilatory demand, and arousal threshold observed from wakefulness to sleep and across sleep stages challenge ventilatory control and can lead to instability in breathing. A structured approach to evaluating breathing instabilities is the loop gain. This engineering term is used to describe the stability of a feedback-controlled system (in this case, the chemical feedback loop controlling an individual's ventilation). In the setting of ventilatory control, loop gain reflects the propensity of an individual to develop periodic (unstable) breathing. Patients with a high loop gain caused by a more sensitive respiratory controller (i.e., high controller gain), more effective CO_2 excretion (i.e., high plant gain), or

$$\text{AHI} = \frac{\text{Apneas (10 sec without flow)} + \text{Hypopneas (reduced flow with desaturation)}}{\text{Total sleep time (h)}}$$

$$\text{RDI} = \frac{\text{Apneas} + \text{Hypopneas} + \text{Arousals (in EEG) associated respiratory event}}{\text{Total sleep time (h)}}$$

Fig. 10.7 Definition of apnea hypopnea index (AHI) and respiratory disturbance index (RDI).

an increased delay because of slowed CO₂ distribution from peripheral tissues to the central chemoreceptors based on, for example, reduced blood circulation (i.e., mixing gain) may be more vulnerable to disturbances of the feedback system, such as slight hypoventilation that occurs as a consequence of decreased upper airway dilator muscle activity during the transition from wakefulness to sleep.

High loop gain may contribute to severity of OSA. Moreover, subjects with a high loop gain have been shown to be more likely to develop breathing instabilities such as Cheyne-Stokes respiration (CSR; see “Central Sleep Apnea”).

SLEEP-DISORDERED BREATHING

SDB refers to the respiratory signs and symptoms associated with sleep-associated respiratory dysfunction and is defined by the occurrence of respiratory events, which are cessations in breathing rhythm (apneas) or momentary or sustained reduction in the amplitude of respiratory flow (hypopnea) during the sleeping state, leading to arterial hypoxemia.⁹⁷ They are usually caused either by increases in upper-airway resistance due to reduction in intraluminal airway diameter (obstructive event), marked reduction or cessation of brainstem respiratory motor output (central event), or both. A specification of SDB into obstructive or central sleep apneas (CSA) is made based on the predominant type of respiratory events. The severity of SDB is usually quantified by the number of respiratory events per hour of sleep. This is necessary because a low number of short apneas and hypopneas can occur up to about five times per hour in healthy subjects. Two measures can be used for this purpose: the apnea hypopnea index (AHI) and the respiratory disturbance index (RDI). Typically the AHI (number of hypopneas and apneas per hour of sleep; Fig. 10.7, top) and RDI (number of hypopneas, apneas, and arousals related to respiratory events per hour of sleep; see Fig. 10.7, bottom) are used to quantify the severity of SDB.

Obstructive Sleep Apnea

Definition. OSA is the most common type of SDB and is diagnosed in patients with more than 15 predominantly obstructive events per hour of sleep, or with fewer respiratory events (i.e., 5–15 per hour) if daytime symptoms (i.e., sleepiness) or comorbidities such as hypertension or atrial fibrillation are present. These cutoff values have been used as an indication for treatment by clinicians based on the clinical guidelines and the international classification of sleep disorders.^{87,88} The severity of sleep apnea is further characterized as mild in patients with less than 15 events per hour of sleep, as moderate with 15 to 30 or less events per hour of sleep, and as severe sleep apnea with 30 or more events per hour of sleep (Table 10.4).^{88,90}

TABLE 10.4 Severity of Sleep Apnea Based on Respiratory Disturbance Index or Apnea Hypopnea Index

	RDI (Events Per Hour)	AHI (Events Per Hour)
No sleep apnea	<5	<5
Mild sleep apnea*	≥5 to <15	≥5 to <15
Moderate sleep apnea	≥15 to <30	≥15 to <30
Severe sleep apnea	≥30	≥30

*Only diagnosed if comorbidities like hypertension, atrial fibrillation of daytime sleepiness are present.

A respiratory disturbance index (RDI) less than 5 per hour is physiologic and found in healthy humans. Sleep apnea is mild with RDI between 5 and 15 per hour, moderate with RDI between 15 and 30 events per hour, and considered severe with an RDI of 30 and greater per hour of sleep. Equal cutoff values are similar for severity based on the apnea hypopnea index (AHI).

Epidemiology. OSA with daytime symptoms affects 0.3% to 5% of the general population.^{98,99} As obesity is one of the major risk factors for OSA, the prevalence might continue to increase with higher rates of obesity in the general population.^{100–102} The prevalence of SDB without daytime symptoms is even higher, with rates of up to 9% in women and 24% in men between the ages of 30 and 60 years, and is often unrecognized.^{103,104}

Furthermore, the prevalence of OSA varies widely between different populations. Individuals with a high susceptibility to OSA include the obese, the elderly, and those with specific comorbidities (e.g., stroke, myocardial infarction). Depending on methodology, the rates of OSA in patients following surgery is between 45%¹⁰⁵ and 75%.¹⁰⁶ A recent study in patients undergoing bariatric surgery revealed an incidence of OSA of as high as 77.5% in this population,¹⁰⁷ while another study showed OSA to be prevalent in 50% of women undergoing gynecologic oncology surgery.¹⁰⁸

Clinical Symptoms. Approximately one-third of patients with OSA complain of typical signs and symptoms occurring during wakefulness. Waking with dry mouth or headache in the morning, daytime sleepiness, falling asleep during monotonous situations (e.g., watching television), as well as subjective impairment of cognitive function are frequently reported. This combination of OSA with daytime symptoms is usually referred to as OSA syndrome.^{109,110}

Sleep-associated signs and symptoms of OSA include witnessed pauses in breathing or snoring, and a high number of nocturnal awakenings, mostly reported as pseudo-nocturia. Patients with OSA occasionally complain of tachycardia

or respiratory distress upon nocturnal awakening (choking sensation) and other symptoms (Box 10.1).

Consequences and Comorbidities. OSA is associated with serious medical consequences, such as hypertension, myocardial infarction, stroke,^{109,111,112} diabetes, diabetic neuropathy, as well as cognitive dysfunction resulting in occupational difficulties and motor vehicle accidents.^{113,114} The development of cognitive impairment in patients with OSA¹¹⁵ is associated with atrophy of brain structures relevant for cognition and memory (hippocampal areas)¹¹⁶ that can be partially reversed by adequate treatment.¹¹⁷ Whether the negative effects of OSA are due to impaired sleep architecture or effects of intermittent hypoxia remain unclear,¹¹⁸ because data investigating the effect of intermittent hypoxia without other symptoms in OSA patients are limited. Nevertheless, recent findings suggest that even asymptomatic OSA might be associated with cardiovascular effects, such as altered daytime autonomic regulation (i.e., heart rate variability).¹¹⁹

BOX 10.1 Symptoms of Obstructive Sleep Apnea

Nighttime Symptoms

- Frequent awakening during the night (e.g., pseudo-nocturia)
- Awakening from own snoring with choking sensation
- Tachycardia
- Sleep that is not restorative

Daytime Symptoms

- Awakening with dry mouth
- Dull headache in the morning
- Daytime sleepiness
- Falling asleep during monotonic situations (e.g., watching television)
- Subjective impairment of cognitive function

Symptoms Reported by Bed Partner

- Snoring, especially when loud and arrhythmic
- Observed pauses in breathing during sleep

Given the frequent rate of complications associated with OSA, it is not surprising that OSA leads to increased socioeconomic costs because of significantly more frequent rates of health-related contacts and medication use, and even higher rates of unemployment.¹²⁰

Risk Factors. Predisposing factors for OSA are obesity,¹²¹ age,^{121,122} male sex,¹²¹ factors leading to swelling of the superficial tissue of the upper airway (smoking),¹²³ allergic rhinitis,¹²⁴ and decreased muscle tone of upper airway dilator muscles (e.g., central nervous system) caused by respiratory depressants.¹²⁵

Pathophysiology. Respiratory events during OSA are characterized by decreased respiratory flow with persisting respiratory effort due to reduced intraluminal diameter of the upper airway up to complete pharyngeal airway collapse (see Fig. 10.6). The muscles involved in respiration are morphologically and functionally skeletal muscles and can be classified into two groups: upper airway dilator muscles and respiratory pump muscles.

Upper airway dilator muscles counterbalance the negative inspiratory pressure generated by the pump muscle to permit airflow during inspiration. Respiratory pump muscles are the collection of muscles responsible for generating inspiratory and expiratory forces in the thorax across the breathing cycle (Fig. 10.8). The patency of the upper airway is maintained by balancing dilating forces (generated by the upper airway dilator muscles) and collapsing forces (i.e., negative intraluminal pressure generated by the respiratory pump during inspiration and compressive extraluminal forces from the surrounding tissues).¹²⁶

UPPER AIRWAY DILATOR MUSCLES. The most extensively studied of the upper airway dilating muscles are the genioglossus and the tensor palatini (Table 10.5). The genioglossus receives a variety of inputs, including phasic (inspiratory) and tonic (noninspiratory) drives, which are distributed differentially across the hypoglossal motoneurons.¹²⁷ In response to negative pharyngeal pressure created by the respiratory pump during inspiration, the

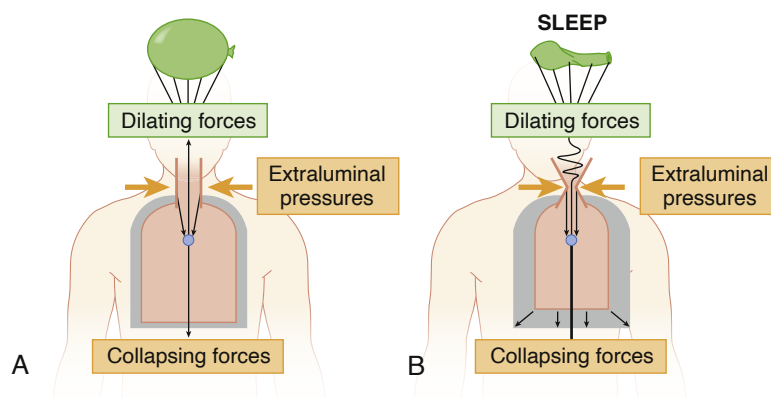


Fig. 10.8 Relationship of upper airway patency and respiratory pump activation. (A) During wakefulness, the force of the upper airway dilator muscles (green balloon, dilating forces) counterbalance the collapsing forces imposed on the upper airway by extraluminal pressures and negative inspiratory pressure generated by the respiratory pump muscles (represented by the orange counterweight [collapsing forces]). In obstructive sleep apnea, (B) sleep onset (blue needle) leads to decreased upper airway patency by reducing the dilating forces. (Modified from Sasaki N, Meyer MJ, Eikermann M. Post-operative respiratory muscle dysfunction: pathophysiology and preventive strategies. *Anesthesiology*. 2013;118:961–978.)

TABLE 10.5 Some Muscles of the Upper Airway Relevant for Upper Airway Stability

Muscle	Tonic Activity	PHASIC ACTIVITY	
		Insp.	Exp.
Tensor palatini ¹⁵³	+	+	–
Levator palatine ¹⁵³	+	+	–
Genioglossus ¹⁵⁴	+	+	–
Geniohyoid ¹⁵⁵	+	–	+
Thyrohyoid	X	X	X

+, Present; –, absent; X, insufficient data. Major muscles of the upper airway and their activity during breathing. Tonic activity, as well as inspiratory (Insp.) and expiratory (Exp.) muscle activity.

genioglossus reflexively stabilizes the upper airway in animals and humans.^{128,129} This reflex is likely a product of signaling from inspiratory modulated motor units. Whereas the genioglossus responds to phasic input on top of its tonic activation, the tensor palatini is considered a tonic muscle with consistent tone throughout the respiratory cycle.¹³⁰

ANATOMIC VULNERABILITY TO COLLAPSING FORCES. The soft tissues of the pharynx are enclosed and stabilized by bony structures, such as the mandible and the spine, and complete collapse of the pharyngeal airway ordinarily requires extraluminal forces, such as hematoma, edema, peripharyngeal masses, or an airway trauma, for instance, as a consequence of prolonged endotracheal intubation.¹³¹ Pharyngeal manifestation of obesity compresses the airway.¹³² Craniofacial abnormalities can further increase the collapsing effects of excessive pharyngeal extraluminal soft tissue in obese patients.¹³³ The extraluminal soft tissue, as well as size and shape of the bony enclosure, are determinants of the extraluminal pressure that need to be antagonized by the upper airway dilator muscle contraction during inspiration, to avoid an upper airway obstruction–related apnea.¹³³ In addition, the upper airway is more vulnerable to collapse in the supine position than in the lateral or sitting positions, due to gravitational effects.^{134,135}

Excessive intravenous fluid administration can affect upper airway patency. In awake, healthy volunteers, the inflation of antishock trousers displaced fluid from the lower extremities to increase neck circumference,¹³⁶ so that the upper airway had a lower threshold to collapse.¹³⁷ This concept is reinforced in studies of subjects with lower extremity venous insufficiency¹³⁸ and congestive heart failure (CHF).¹³⁹ These studies have shown nocturnal redistribution of fluid from the lower extremities into the neck increases upper airway collapsibility¹³⁸ and the severity of central apnea and OSA.¹³⁹ Similar effects may increase the risk for OSA during pregnancy and early after delivery.¹³⁹

Another important component of airway patency is the interplay between lung volume and upper airway collapsibility. Higher end-expiratory lung volumes are associated with a decrease in upper airway resistance to airflow in awake healthy humans,¹⁴⁰ and an increase in upper airway lumen dimensions in subjects with and without OSA.¹⁴¹ The mechanism for the interaction between upper airway patency and lung volume is thought to lie in the generation of longitudinal traction forces in the trachea.^{142,143} Upon inspiration, the

lung inflates and effectively forces the carina into a more caudal position, creating stretching forces on the fixed trachea.¹⁴² These forces are transferred to the side walls of the upper airway.¹⁴³ Effectively, tracheal traction allows the respiratory pump muscles to contribute to upper airway opening.

COLLAPSING INTRALUMINAL PRESSURE. The respiratory pump is the motorized action driving inspiration and expiration. It is the force that broadens the thoracic cavity and creates negative intrathoracic pressure to draw a breath in, and when needed, positive intrathoracic pressure to exhale rapidly. Inspiratory pump muscles are an anatomically diverse group, with the most studied being the intercostal muscles and the diaphragm. The diaphragm accounts for 60% to 70% of lung volume change during quiet respiration.¹⁴⁴ During inspiration, the volume of the thoracic cavity increases as a result of contraction of the diaphragm and the external intercostal muscles. The lungs expand secondarily to the increased negative intrathoracic pressure generated by these phasic actions of the inspiratory pump muscles. The negative intrathoracic pressure translates into a negative intraluminal pressure in the upper airway, forcing the airway to collapse as soon as this pressure drops below a critical value.¹⁴⁵

Although this critical airway pressure (Pcrit) is typically negative (approximately -5 cm H₂O) in healthy controls, the upper airway in patients with OSA during sleep may even collapse at a positive Pcrit. Positive intraluminal pressure is required for reopening the paralyzed airway in patients with OSA.¹⁴⁶ Under normal conditions, the two compressing factors, (1) negative intraluminal pressure and (2) positive extraluminal pressure, need to be actively compensated by muscular activity of the upper airway dilator muscles to maintain airway patency.⁹⁷

WAKEFULNESS STIMULUS AND SLEEP. Excitatory inputs to the upper airway motoneurons (e.g., hypoglossal motoneurons) include those from serotonergic and noradrenergic neurons that fire predominantly during wakefulness,^{147–150} resulting in a “wakefulness stimulus” that increases the activity of upper airway dilator muscles during this arousal state. With onset of sleep, this arousal-dependent neuronal input (active while awake versus less active, or inactive when asleep) disappears, leading to a decrease in upper airway muscle activity and causing increased upper airway resistance in healthy controls and airway collapse in patients with OSA.^{151–153} The resulting retropalatal obstruction of the upper airway is reported to be the most common pathophysiological mechanism in OSA.¹⁴⁶

ONE WAY VALVES IN THE UPPER AIRWAY. Observing the respiratory flow tracings in patients with OSA, one can find some respiratory events that do not show the characteristic progressive flow limitation ultimately resulting in complete upper airway occlusion, typically seen as a result of “collapsible tube pathophysiology” with variable compliance. In these cases, the flow tracings show normal expiratory flow and sudden reduction/cessation before reaching maximum inspiratory flow. This “rapid obstruction” only during the inspiratory phase occurred in approximately 20% to 30% of OSA patients.¹⁵⁴ Recent findings suggest that there are two one-way valves in the pharynx. While the epiglottic valve closes during inspiration, the soft palate may function as a second one-way valve limiting flow during expiration.¹⁵⁵

Similar to the epiglottis, the soft palate hangs from the hard palate like a peninsula in the oropharynx and might

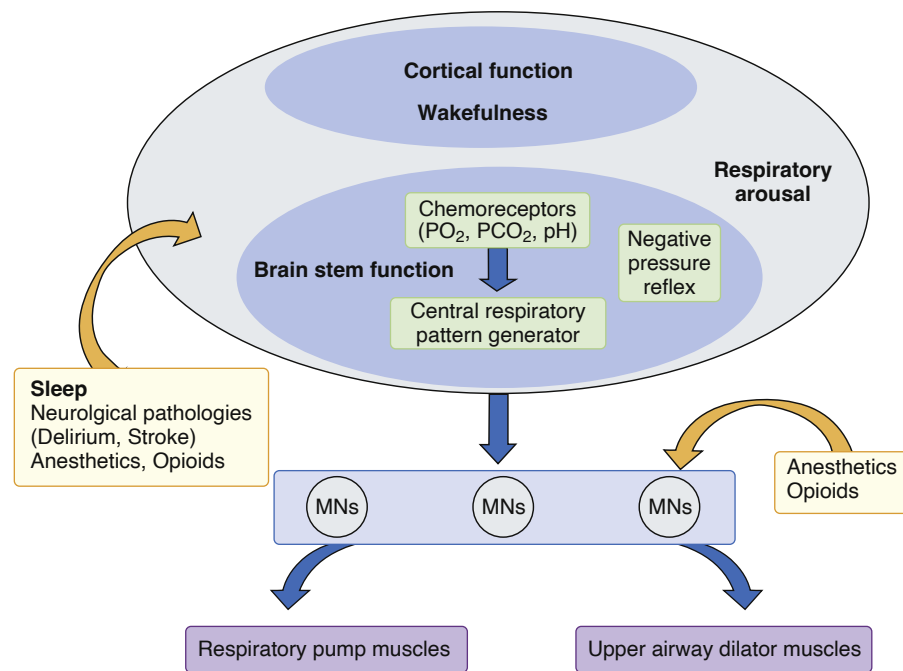


Fig. 10.9 Effects of respiratory arousal on upper airway dilator and respiratory pump muscles. Respiratory arousal is composed of three primary inputs: central respiratory pattern generator processing peripheral and central chemoreceptor afferents, reflex responsiveness to the magnitude of negative pressure in the airway generated by the respiratory pump muscles, and strength of the wakefulness stimulus. Different factors can impair respiratory arousal, such as sleep and neurologic pathologies, anesthetics, and opioids. Blue arrows indicate excitatory effect; yellow arrows indicate inhibitory effect. *MN*, Motor neuron. (Modified from Sasaki N, Meyer MJ, Eikermann M. Postoperative respiratory muscle dysfunction: pathophysiology and preventive strategies. *Anesthesiology*. 2013;118:961–978.)

occlude the upper airway during expiration.¹⁵⁶ A recent study found that sudden reduction of expiratory airflow immediately after peak expiratory flow occurs more often in OSA patients under positive pressure ventilation.¹⁵⁷ Similar expiratory flow limitation pattern was also reported in spontaneously breathing OSA patients, suggesting that this phenomenon is not unique to positive pressure ventilation.¹⁵⁸

These findings may point to a second, slightly different mechanism underlying OSA than the pure collapsible tube pathophysiology in some OSA patients, which should be particularly important during anesthesia. Correct identification of the closure site in each OSA patient will enable the clinician to provide an individualized OSA treatment.¹⁵⁵

RESPIRATORY AROUSAL. Respiratory arousal is arousal from sleep owing to cumulative and progressive increases in stimuli related to breathing (hypoxia, hypercapnia, and respiratory effort).¹⁵⁹

Three primary inputs contribute to arousal-related restoration of breathing following an apnea (Fig. 10.9):

1. Peripheral and central chemoreceptors sensitive to partial pressures of oxygen and carbon dioxide¹⁶⁰
2. Sensors in the upper airway responsive to negative pressure generated by the respiratory pump^{128,129}
3. Cortical inputs directly related to state of consciousness or wakefulness¹⁶¹

Any of these inputs can restore respiratory muscle tone if the magnitude of stimulation is sufficient. Cortical awakening from sleep, identified by EEG signs of wakefulness, is an adequate stimulus for ventilation. However, obstructive apneas, such as upper airway collapse in OSA, can be terminated by an increased drive to the respiratory muscles

not involving cortical arousal.¹⁶² For example, hypercarbia resulting from sustained hypopnea¹⁶⁰ and elevated upper airway negative pressure^{128,129} can independently restore tone to the respiratory muscles. The level of drive provided to the respiratory muscles depends on the summation of stimuli in the central respiratory pattern generator output, including peripheral and central chemo-responsiveness, reflex responsiveness to the negative airway pressure, and strength of the wakefulness drive.

Treatment. Adequate treatment of OSA improves nocturnal oxygen saturation as well as duration and quality of sleep, which translates to reduced daytime sleepiness, improvement of daytime functions, and improved quality of life. Successful treatment of OSA reduces cardiovascular risk, improves insulin sensitivity, and increases neurobehavioral performance.^{163–165} Therefore, all patients with a diagnosis of OSA should be offered treatment as soon as the diagnosis of OSA has been established and its severity has been determined by objective testing (i.e., PSG).⁸⁸ Although different treatment options for OSA have been developed during the last decades (Table 10.6), continuous positive airway pressure (CPAP) dose dependently increases upper airway diameter and is still the most effective treatment for OSA of all severities.^{166–168} However, a recent study has questioned if CPAP treatment is able to prevent the long-term adverse cardiovascular events in OSA.¹⁶⁹

POSITIVE AIRWAY PRESSURE TREATMENT. CPAP as a treatment for SDB is typically applied continuously with a nasal or oronasal mask, and can dose dependently reverse any sleep-associated upper airway obstruction, as outlined in Fig. 10.10. The CPAP level needed for treatment of OSA

TABLE 10.6 Treatment Possibilities for Obstructive Sleep Apnea			
Treatment	Procedure/Device	Recommendation	References
Weight reduction	Reduction of body weight Weight-loss surgery (improves success of weight loss)	Medium to high, SU	170,171
Medication	Drug-based treatment (e.g., tricyclic antidepressants, serotonin reuptake inhibitors, cholinergic agonists, carbonic anhydrase inhibitors)	NR, ID	90
Surgical	Nasal surgery Palatal surgery and implants Tongue base surgery	Low, SU Low, MC Low, MC	172,173
Increase of muscle activity	Muscle training Hypoglossal nerve stimulation	ID ID	174
Nonsurgical	Oral appliance Positive airway pressure	High, AT High, GS	175 175

AT, Alternative treatment if positive airway pressure is not tolerated; GS, gold standard; ID, data not sufficient for recommendation; MC, may be concerned in carefully selected patients if conservative treatment fails; NR, not recommended; SU, supportive treatment.

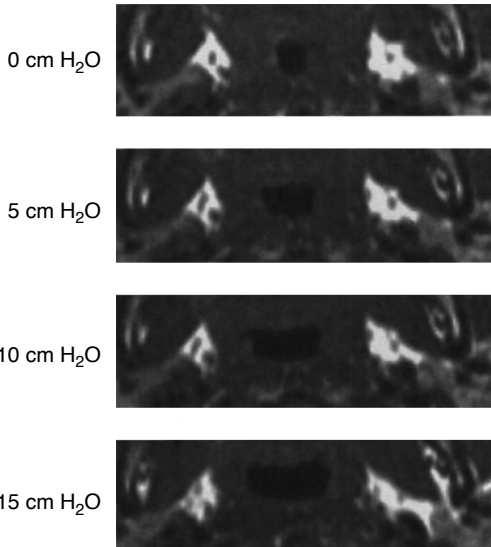


Fig. 10.10 Magnetic resonance imaging of the human upper airway during different levels of continuous positive airway pressure. Magnetic resonance imaging of the upper airway during continuous positive airway pressure of 0, 5, 10, and 15 cm H₂O shows a dose-dependent increase in upper airway diameter. (Obtained from Schwab RJ, Pack AI, Gupta KB, et al. Upper airway and soft tissue structural changes induced by CPAP in normal subjects. *Am J Respir Crit Care Med.* 1996;154[4 pt 1]:1106–1116.)

varies mostly between 5 and 20 cm H₂O. However, home RP and titration of CPAP might be similarly effective in some but not all patients with suspected OSA.^{176,177} After titration, the prescribed treatment pressure is applied continuously throughout the night. Although this treatment is efficient in reversing the underlying pathology, the impact of CPAP on outcome is limited by patient adherence,^{112,178} mostly because of local side-effects at the nose or face, or discomfort caused by the mask.¹⁷⁹ When applied with high pressure, the amount of airflow can prevent the patient from falling asleep while using the treatment device. Some treatment devices offer a ramp or delay function that gradually increases treatment pressure from

a low starting pressure to the prescribed pressure over a period of 5 to 45 minutes, allowing the patient to fall asleep more easily. Some patients report difficulties exhaling against high CPAP. To avoid this problem, bilevel treatment with reduced expiratory positive airway pressure and a sufficiently high inspiratory positive airway pressure can be used.

In some cases, one CPAP pressure level is not sufficient to treat sleep apnea. CPAP devices with dynamic pressure levels can improve treatment success, particularly in patients with a variable severity of sleep disordered breathing during different sleep stages. These automatic positive-airway pressure or auto-titrating devices measure different variables associated with hypopnea, such as oropharyngeal wall vibration, snoring, and inspiratory flow limitation, and increase the airway pressure until these signs and symptoms of hypopnea disappear. In addition, it is not entirely clear whether CPAP treatment is able to prevent the increased cardiovascular risk inherited by OSA.

CPAP is sufficient in the treatment of most patients with OSA. However, some patients may need different treatment methods. For example, patients with mixed apneas (obstructive and central) or predominantly central apneas need a more controlled (frequency or time controlled), noninvasive ventilation (NIV) with predefined minimal respiratory frequency or respiratory timing that automatically induces the next inspiration (ventilation) if the patient does not induce the next breath within preset parameters.

ALTERNATIVE TREATMENT OPTIONS. Oral appliances (OAs) are possible treatment options in patients with mild to moderate OSA who do not tolerate CPAP therapy. Two major designs are currently used clinically: (1) mandibular repositioning appliance, which holds the mandible in an advanced position (a protrusion of at least 50% of the maximal possible extension is recommended for effective treatment),¹⁷⁹ and (2) tongue-retaining devices, leading to repositioning of the tongue into a forward position without any protrusion of the mandible. A multidisciplinary

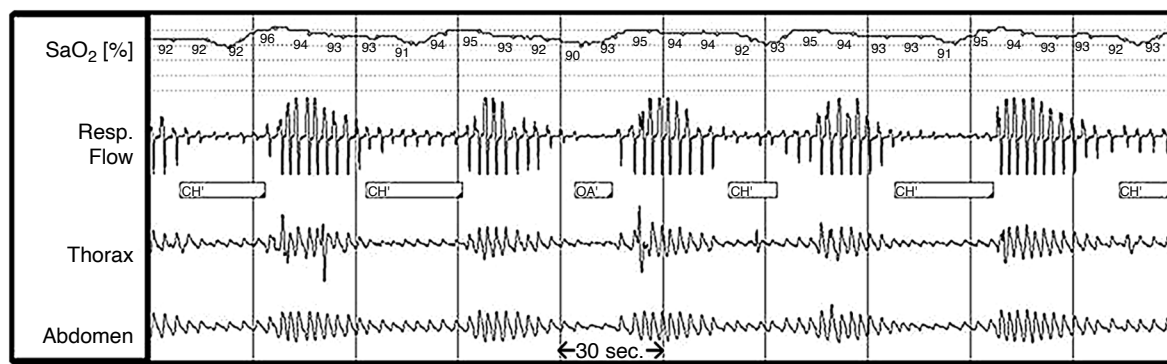


Fig. 10.11 Polysomnographic recording of an episode of Cheyne-Stokes respiration. CH, Central hypopnea; OA, obstructive apnea.

approach including a sleep physician and a dental technician experienced in OAs has been recommended for this purpose, because both factors are crucial for patient adherence and outcome from OA treatment.¹⁰⁹ OAs are recommended in patients with mild to moderate OSA who do not tolerate,¹⁷⁹ do not respond to, fail, or are not appropriate candidates for CPAP treatment.⁸⁸

Historically, surgical treatment methods were the only approach to treatment of OSA, but the effectiveness of most of the nasopharyngeal surgical approaches for severe OSA remains unclear. However, tonsillectomy is beneficial in adult patients with OSA from tonsillar hypertrophy, and adenotonsillectomy is recommended in children with OSA and adenotonsillar hypertrophy.¹⁸⁰ Repeated sleep testing before and following surgery is recommended to ensure sufficient long-lasting therapeutic effects.^{88,179,181}

Electrical stimulation of the upper airway muscles has been explored recently as a new approach to treatment of OSA. Hypoglossal nerve stimulation induces genioglossus muscle contractions, which dose dependently increase inspiratory airflow during OSA,¹⁸² but the invasiveness of the procedure, the cost of the device, and the requirement for preimplantation endoscopy during anesthesia limit its use as a first-line treatment of OSA.¹⁴⁹

While pharyngeal fat deposits led to a decrease in pharyngeal patency, underlining the risk factor of obesity,^{183,184} weight loss leads to a reduction in Pcrit and the severity of OSA^{145,171} and is recommended as adjunctive treatment in all overweight patients with OSA.¹⁸⁵ Because long-term weight reduction is more effective when accompanied by weight loss surgery, bariatric surgery may be considered as an adjunctive therapy in very obese patients (body mass index [BMI] ≥ 40 kg/m²), as well as in those with important comorbidities and BMI of 35 kg/m² or greater and in whom dietary attempts at weight control have been ineffective.⁸⁸

Oxygen is currently not recommended as primary treatment of OSA, but may be used as an adjunctive treatment in some patients, particularly in the postoperative period.^{88,186}

Although OSA is more common in the older population, it occurs in children with a peak in the incidence between 2 and 5 years of age. Obesity predicts snoring and other signs of obstructive respiratory events in children.¹⁰⁹ Tonsillar and adenoidal hypertrophy is another important cause of OSA in children and can be treated surgically.¹⁸⁷

Central Sleep Apnea

CSA impairs quality of life¹⁸⁸ and is associated with adverse outcome in heart failure patients.^{189,190} CSA is defined as cessation of air flow without respiratory effort,¹⁸⁸ which separates it from OSA, where respiratory effort is maintained or even increased during an apnea. In clinical sleep apnea, considerable overlap between OSA and CSA occur that need to be identified and subsequently treated.¹⁹¹

CSA can be found in older patients and in patients with severe comorbidities such as CHF, stroke, or other neurologic disorders (e.g., amyotrophic lateral sclerosis). In a southern Pennsylvania cohort, CSA (AHI ≥ 20 /hour) was found in 5% of men aged 65 years and older, but it was not found in younger men or in women of any age. For an AHI of 2.5 per hour and greater, the prevalence estimates in men younger than 45 years remains negligible, while CSA was found in 1.7% of men between 45 and 64 years, and 12% in men older than 65 years,¹⁹² or 9% in individuals between 40 and 97 years, respectively.¹⁹³

CSA mechanisms can be categorized into those with high and low loop gain. The most common subtype of CSA with an increased loop gain is CSR and is commonly seen in patients with CHF and left-ventricular systolic dysfunction. CSR is defined as a crescendo-decrescendo pattern of hyperventilation between 20 and 30 seconds in duration, followed by 10 to 40 seconds of hypopneas or apneas (Fig. 10.11), usually occurring during NREM sleep stage 1 and 2.¹⁹⁴ Nevertheless, CSR can also occur during exercise or wakefulness. Nearly one out of two patients with CHF shows CSR.¹⁹⁵ CSR is more common in men and worse in the supine body position.¹⁶¹

Potential respiratory therapies of CSR include oxygen, respiratory stimulants (e.g., CO₂, theophylline, and acetazolamide), and NIV such as bilevel positive airway pressure. The effectiveness of CPAP as therapy is controversial. Optimization of medical therapy is the best treatment, because CSR will often resolve with adequate treatment of CHF (cardiac resynchronization therapy and surgical treatment, such as heart transplant).¹⁹⁴

Other Forms of Central Breathing Disorders. The term *periodic breathing* refers to altitude-induced breathing instability that occurs in subjects transferred to a high altitude, where ambient hypoxia caused by low barometric pressure leads to an increase in controller gain.⁹⁷ Idiopathic CSA,

which is a relatively uncommon disorder at sea level, occurs more likely in individuals with an elevated hypercapnic ventilatory response (high controller gain) that leads to hypocapnia and respiratory control instability during sleep. Patients with idiopathic CSA tend to have low PaCO_2 levels, even during wakefulness.¹⁹⁶

Obesity Hypoventilation Syndrome

Alveolar hypoventilation is defined as insufficient ventilation leading to hypercapnia (increased PaCO_2). Mechanisms of alveolar hypoventilation include central hypoventilation, chest wall deformities, neuromuscular disorders, chronic obstructive pulmonary disease, as well as severe obesity (obesity hypoventilation syndrome [OHS]). OHS is defined as the combination of nocturnal and daytime hypoventilation, usually leading to hypercapnia in obese subjects ($\text{BMI} \geq 30 \text{ kg/m}^2$), in the absence of other causes of hypoventilation.

The prevalence of OHS is estimated to be up to 50% in obese patients with OSA. It can occur in up to 50% of patients with a BMI of 50 kg/m^2 or more,¹⁹⁷ compared with 0.15% to 0.3% in the general adult population. Ninety percent of patients with OHS also suffer from OSA.^{198,199} OHS often remains undiagnosed, and the true prevalence remains unclear.

Severe obesity is associated with an increase in respiratory drive that helps maintain eucapnia in the presence of the abnormal chest wall mechanics and high work of breathing.^{200,201} In OHS, this compensatory mechanism is abolished,^{201,202} which in part might be explained by leptin resistance.^{203–205} Typically, OHS manifests as a reduced lung capacity, vital and functional residual capacity, expiratory reserve volume, respiratory system compliance, and inspiratory muscle strength, whereas response to CO_2 might be reduced or normal. In addition, there is an increase in serum bicarbonate and alveolar PCO_2 , as well as in the work of breathing and leptin levels.^{205,206}

Impairment of the effectiveness of respiratory pump muscle function in patients with OHS can be explained by the effects of low lung volume because of central fat distribution, leading to a cranial displacement of the diaphragm when lying supine.^{206,207} In addition, diaphragmatic myopathy might be a contributing factor of OHS.²⁰⁵ Treatment options include weight loss and NIV.^{207,208}

Sleep and Anesthesia: Two Unequal Twins Influencing Perioperative Medicine

CLINICAL PICTURE OF SLEEP AND ANESTHESIA

Although physiologic sleep and anesthesia share some clinical features (loss of consciousness and a modulation of brainstem autonomic function), major differences can be found when closer observations of both behavioral states are undertaken. In contrast to anesthesia, sleep shows spontaneous generation and termination, ready reversibility by noxious stimuli, and homeostatic regulation. Anesthesia does not share the stage-wise structure seen during physiologic sleep. In addition, functional imaging studies underline the fundamental differences during onset of anesthesia and wakefulness-sleep transition.

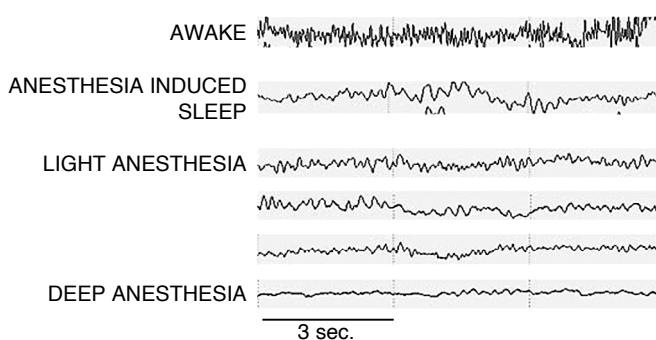


Fig. 10.12 Electroencephalographic recording during sedation and loss of consciousness induced by propofol. Compared with physiologic sleep (see Fig. 10.1), sedation shows electroencephalographic (EEG) traces comparable to slow wave sleep, indicating the sleep-inducing effect of anesthetics. Anesthesia-induced loss of consciousness (lower four traces), shows different EEG activity with reduced amplitudes and burst suppression. (Burst suppression is not shown in these traces.)

Although there are similarities in the EEG patterns between slow-wave sleep and anesthesia-induced unconsciousness, the EEG patterns during sleep and anesthesia are different,²⁰⁹ with different frequencies and types of activation in EEG recorded during physiologic sleep and the uniform EEG picture observed during anesthesia-induced unconsciousness (Fig. 10.12).²¹⁰

During anesthesia induction, the level of consciousness continuously decreases from fully awake through states of reduced response to external stimuli up to a completely unresponsive state.²¹¹ This is in contrast to the sharp α - θ EEG transitions seen while an individual proceeds from wakefulness to sleep without any transient stages. Individuals in a steady state of physiologic sleep can be aroused with sufficient stimulation, whereas drug-induced unconsciousness needs at least some drug elimination to occur before individuals can be aroused.²¹²

ACTIVATION OF SLEEP-PROMOTING PATHWAYS DURING ANESTHESIA

The role of the endogenous sleep-promoting system in the mechanism of action of general anesthetics has recently received much attention. This hypothesis is attractive because of the similarities of sleep and anesthesia and some evidence that anesthetically induced sleep could meet some of the homeostatic need for sleep.^{213,214} It is even more attractive because, despite the large database of the molecular effects of anesthetics, how loss of consciousness occurs is still not explained. An important discovery is that some of the neurons that are active during sleep in the VLPO are also activated by certain anesthetics.^{213–216} A second key observation is the inhibition of the arousal nuclei, the TMN, contributes to anesthesia.²¹⁵ This effect also implicates activation of VLPO because it provides the major source of inhibition of the TMN. The idea has emerged that anesthesia-induced loss of consciousness is mediated by effects of the anesthetics on the flip-flop switch centered in the VLPO that is responsible for the sharp transition from wakefulness to sleep and back (see earlier section).^{212,217,218} However, there are a number of problems with this theory, including that rats and mice with complete lesions of the VLPO can still be anesthetized.^{216,219} Although lesions of VLPO lead to a

transient resistance to volatile anesthetics,²¹⁶ if some time elapses following a VLPO lesion, the animals show increased sensitivity to isoflurane anesthesia attributable to increased homeostatic sleep drive.^{216,219-221} Another problem with the VLPO-TMN circuit hypothesis of anesthesia is that direct inhibition of the TMN produces sedation but not anesthesia.²¹⁵ Animals without VLPO neurons have profound insomnia. Therefore, interaction of anesthetics with these sleep-promoting nuclei alone is not sufficient to produce unconsciousness during anesthesia. During recent years, several studies provided evidence to extend the scientific framework of general anesthetics modulating sleep-promoting regions by the anesthetic depression of arousal-promoting nuclei, including the LC, GABAergic neurons of the pontine reticular formation, the pedunculopontine and laterodorsal tegmentum, the ventral tegmental area, as well as the perifornical area, the tuberomammillary nucleus, and the basal forebrain.^{222,223} This “bottom-up” hypothesis of anesthesia-induced unconsciousness may be the result of the simultaneous interaction with the brainstem and diencephalic regions, including the AAS and endogenous sleep circuitry. Although, these mechanisms show similarities with sleep-promoting neuronal interactions, they are unlikely to be the only correlate of anesthesia-induced unconsciousness. Sleep, but not anesthesia, is easily reversed or prevented by real or perceived environmental threats. While interaction with sleep-promoting neurons work by inhibiting these arousal pathways, other neuronal networks crucial for consciousness (e.g., thalamocortical networks) remain active during sleep.²²⁴ However, the finding that many anesthetics engage sleep pathways is still important and meaningful, as this is undoubtedly an important mechanism by which these classes of anesthetics promote sleep. Nevertheless, this effect does not and cannot prevent arousal to sufficient external stimuli. The unique property of anesthesia to do so has to be mediated by an additional mechanism.

PERIOPERATIVE INTERACTIONS BETWEEN ANESTHESIA AND SLEEP

Anesthesia and painful surgical procedures affect sleep and circadian rhythms²²⁵⁻²²⁷ for as long as 6 months, depending on the complexity of the procedure performed.²²⁸ A substantial decrease in REM sleep occurs on the first night after surgery and anesthesia, followed by a profound REM rebound phenomenon on the second to fourth postoperative nights, when REM sleep increases in both intensity and amount.^{225,227} Although most anesthetics can lead to an impairment of sleep architecture, such as REM depression and reduced sleep quality during the early postoperative period, the extent of this impairment probably depends on the pharmacokinetics and pharmacodynamics of the anesthetics and opioids used,^{229,230} which affect the duration of REM sleep,^{231,232} as well as surgery-induced stress.

The effect of propofol on sleep architecture and REM sleep is complex and dose dependent. In long-term ventilated, critically ill patients, propofol sedation abolishes REM sleep and diminishes sleep quality,²³³ whereas with low-dose propofol, REM sleep is possible. The effects of ketamine sedation on sleep architecture have not been studied in detail, although it may have rather mild effects on REM duration.²³⁴

Anesthetics have strong effects on GABA and NMDA receptors, both of which are linked with circadian control. As a result, anesthetics may interfere with entrainment of the circadian rhythm; however, preclinical and human data show inconsistent results. A recent trial in honeybees suggested an anesthetically induced circadian shift. Inside the laboratory with no outdoor cues, daytime anesthesia disturbed the usual activity patterns in the hive for the next several days. In addition, cycles of clock gene activity showed a delay,^{235,236} and inhibition of the period circadian protein (per)-2 gene expression (one of the core clock genes) was observed during anesthesia.²³⁷ Of note, psychotropic nonanesthetics, such as opioids, have been shown to affect melatonin secretion indirectly, independent of the behavioral state anesthesia in a pig model.²³⁸ Previous work in humans has shown that 3 hours of anesthetic exposure does not affect the circadian phase of the body temperature rhythm.²³⁹

In summary, the specific mechanisms of anesthetics effects on circadian rhythm are unclear. The ability to distinguish between effects occurring directly on the circadian pacemaker and those occurring downstream from the pacemaker on other physiologic control systems requires additional research under rigorous experimental conditions.

Surgical treatment itself impairs sleep, expressed as reduced REM sleep duration even in the absence of general anesthesia.^{229,230} Pain, inflammation, stress immobility, and anxiety seem to be contributing factors.²⁴⁰ After surgery, patients show a significant impairment of sleep quality and duration. Eventually, it will be known whether specific anesthetics are preferable to avoid the adverse effects of sleep deprivation.

SLEEP-DISORDERED BREATHING AND AIRWAY PATENCY DURING ANESTHESIA

Perioperative complications occur more often in patients with OSA than in healthy controls,^{241,242} but the reasons for this observation are unclear. It is challenging to isolate the effect of OSA from the known adverse effects on perioperative outcome of typical OSA comorbidities, such as hypertension, diabetes, coronary artery disease, neurovascular vulnerability, and obesity. Fortunately, severe postoperative complications are uncommon. As a consequence, large trials are needed to isolate the causal relationship between SDB and perioperative complications, such as severe respiratory failure, thromboembolic complications, increased hospital length of stay, and mortality.

Independent of obesity, OSA is not associated with difficult tracheal intubation or mask ventilation.²⁴³⁻²⁴⁵ However, recent metaanalyses reported a significantly higher risk of postoperative major cardiac and cerebrovascular events, newly detected postoperative atrial fibrillation and acute postoperative respiratory failure, compared with control subjects.²⁴⁶⁻²⁴⁸ These findings need to be confirmed in future prospective studies with a larger sample size and a more homogenous population such that confounding factors can be controlled. OSA is associated with postoperative delirium,^{241,249} which is an important perioperative complication associated with increased costs, morbidity, and mortality. However, it is currently unclear if this is due to repetitive hypoxia or sleep fragmentation. In contrast, a

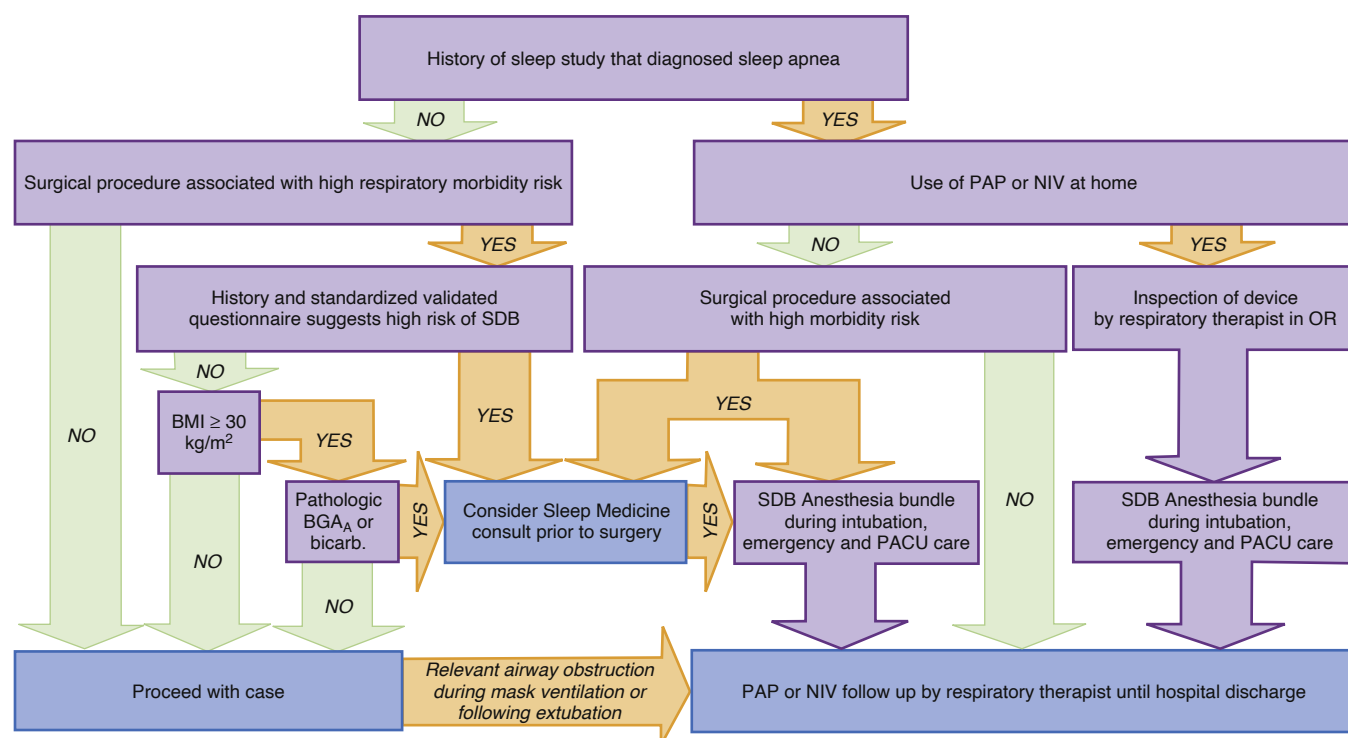


Fig 10.13 Clinical pathway for perioperative management of patients with sleep-disordered breathing. *BGA_A*, Arterial blood gas analysis; *bicarb*, venous bicarbonate level; *BMI*, body mass index; *NIV*, noninvasive ventilation; *SDB*, sleep-disordered breathing; *PACU*, postanesthesia care unit; *PAP*, positive airway pressure.

recent study using the Nationwide Inpatient Sample database analyzed data from 1,058,710 patients undergoing elective surgery. These authors showed that patients previously diagnosed with SDB had significantly higher rates of emergent mechanical ventilation, use of NIV, and CPAP, as well as increased rates of respiratory failure. Nevertheless, the study also showed that patients with SDB had better outcomes after emergent intubation and had lower care-associated costs compared with their non-SDB counterparts.²⁵⁰ Similar findings were reported by the same group for data from 91,028 patients undergoing bariatric surgery.²⁵¹ In this study, patients with a previous diagnosis of SDB had a shorter hospital length of stay and lower total costs compared with non-SDB patients. The reasons for the differences are unclear. Patients with a known history of sleep apnea may receive more sophisticated perioperative monitoring and more aggressive treatment of their respiratory disease, which might also explain the higher rate of postoperative reintubation in SDB patients reported in the same group.²⁵⁰ Another possible explanation relates to the consequences of chronic nocturnal desaturation, which can have preventive effects in terms of the consequences of perioperative acute hypoxia. Most importantly, these findings indicate that the association of SDB, and perioperative outcome is likely to be more than a unidirectional relationship. More likely is that SDB could have a dual effect, leading to increased rates of perioperative respiratory complications in patients with SDB, but additionally protecting the same population from the fatal consequences of these complications. The mechanism of the potential link between OSA and postoperative respiratory failure, as well as postoperative delirium, is probably multifactorial. Frequent episodes

of airway collapse in OSA lead to hypoxia, disrupted sleep, daytime sleepiness, and increased arousal threshold from sleep, all of which can be considered as potential precipitating or augmenting factors of postoperative complications. It is intriguing to note that patients with OSA appear to be most vulnerable to hypoxia on postoperative night two or three, the time interval during which postoperative delirium most typically manifests.

One population that might deserve additional attention in perioperative care is patients with OHS who, compared with obese patients, are more likely to be admitted to a hospital and require more healthcare resources.^{250,252} In addition, higher rates of ICU admission, lengthier long-term care requirements at discharge, and higher mechanical ventilation rates have been reported in patients with OHS compared with severe obese patients without OHS.²⁵¹⁻²⁵⁴

In summary, patients with SDB are more vulnerable to developing severe perioperative complications. Nevertheless, it is currently unclear whether these complications translate to a poorer overall postoperative outcome for these patients.

PERIOPERATIVE MANAGEMENT IN PATIENTS WITH SLEEP-DISORDERED BREATHING

The standard of care for perioperative management of patients with SDB depends on the severity of the disease, comorbidities, and the threat associated with the planned surgical procedure. It is probably neither feasible nor necessary to conduct a sleep study in all obese patients at risk for sleep apnea who are scheduled for surgery; however, high-risk patients need to be identified and treated perioperatively. Fig. 10.13 shows an algorithm that is being

developed by an interdisciplinary international group of clinicians. The algorithm is driven by the idea that patients who have been prescribed CPAP preoperatively should continue to use CPAP perioperatively. Patients who carry several risk factors (procedural and comorbidity-based) may need a preoperative sleep medicine consultation and/or a CPAP prescription.

Preoperative Screening

Several strategies need to be considered to identify patients with SDB adequately.^{94,253,254} Although clinical examination is the easiest and most cost-effective method of assessment using predisposing physical characteristics, it has a poor sensitivity and specificity of only 50% to 60% to diagnose OSA accurately.^{255,256} However, the clinical value of using questionnaires relates to the fact that they help to identify a high-risk perioperative population, independently of the question of whether an individual patient has OSA. For example, the STOP-Bang questionnaire basically screens for comorbidities of OSA that we know increase the risk of perioperative complications (hypertension, obesity, male sex, older age) rather than direct breathing disorder-associated characteristics. The Score for Prediction of Obstructive Sleep Apnea might also be a helpful tool to predict patients with OSA and its perioperative outcome based on comorbidities and easily available clinical data.²⁵⁷

PSG is required to identify SDB specifically, but it cannot be used for preoperative screening of surgical patients. PSG is expensive, may delay surgery, and is inconvenient for the patient. A multistep approach of preoperative screening is required. Preoperative assessment should include a screening for SDB and for the current use of NIV. OSA might not represent real risk in patients scheduled for low-risk procedures from a surgery and anesthesia point of view. The authors believe that patients with no previously diagnosed SDB undergoing a low-risk surgical procedure should receive standard perioperative monitoring and treatment.

Patients undergoing high-risk surgical procedures should receive further workup using a clinical examination combined with standardized and validated questionnaires (e.g., Berlin).^{254,258} Although this diagnostic algorithm might be a sufficient screening for OSA, it might be insufficient to detect OHS.^{251,253} A blood gas analysis should be considered as a method to detect hypercapnia as a major symptom of OHS (criterion: awake daytime hypercapnia $\text{PaCO}_2 \geq 45$ mm Hg). Alternatively, a venous bicarbonate concentration ≥ 27 mmol/L is highly sensitive (92%) for an increased PaCO_2 , which combined with hypoxemia ($\text{SpO}_2 \leq 94\%$) indicates a high risk of OHS in these patients.^{251,253,256,259} Of note, OHS is a diagnosis by exclusion. Severe obstructive airway diseases, severe interstitial lung disease, severe chest wall disorders (e.g., kyphoscoliosis), severe hypothyroidism, neuromuscular disease, and congenital central hypoventilation syndrome need to be excluded, and further evaluation of sleep by a sleep medicine specialist before anesthesia should be considered. In these cases, the sleep specialist should determine the optimal diagnostic instruments in cooperation with the perioperative team, prescribe the perioperative auto-titrating CPAP device, and offer in collaboration with respiratory therapy specialists to improve tolerability of perioperative CPAP therapy.

Perioperative Management of Patients with Possible Sleep-Disordered Breathing

Although the data on optimal intraoperative and perioperative management are still limited, and mainly based on studies with low sample size, the techniques and procedures summarized in [Box 10.2](#) might be considered during the perioperative management of patients with SDB.²⁵⁸ Patients with a history of SDB who are undergoing surgery and anesthesia with high risk of morbidity should receive an “OSA anesthesia bundle” (see [Box 10.2](#)) that includes special procedures and preparations during tracheal intubation, extubation, pain therapy, and perioperative CPAP therapy.

Tracheal Intubation. Patients with OSA are frequently obese, and obesity is a risk factor for difficult intubation.^{244,259-261} Patients should be preoxygenated efficiently before injection of the anesthetic drug. Sniffing and reverse Trendelenburg position are preferred before intubation, which improves maintenance of the passive (i.e., paralyzed) pharyngeal airway in patients with OSA^{260,262} and increases functional residual capacity. To reduce the duration of paralysis, nondepolarizing neuromuscular blocking drugs should be used with caution supported by quantitative neuromuscular transmission monitoring. Immediately after successful intubation, a lung recruitment maneuver and the application of positive end-expiratory pressure for maintaining lung volume during surgery should be considered.^{261,263}

Intraoperative Treatment. Impaired upper airway patency is of special concern whenever anesthetics are administered without a device that bypasses the collapsible upper airway—for example, during sedation for endoscopy.^{264,265} While the overall rate of overt sedation-related complications for such procedures (e.g., endoscopy) appears low,²⁶⁶ patients who are vulnerable to respiratory complications may suffer from sequelae of increased upper airway collapsibility beyond the immediately observable after intervention.

Opioids and GABAergic sedatives/hypnotics impair upper airway patency without any meaningful difference between compounds given at an equipotent dose of analgesics.²⁶⁷⁻²⁷⁰ Of note, the impairing effects of GABAergic drugs on airway patency can be reversed by the respiratory stimulants ([Fig. 10.14](#)).²⁶⁶ Thus it is recommended not to extubate patients at risk of postoperative airway collapse during hypocarbia but rather at mildly hypercarbic conditions.

Ketamine abolishes the impairment of upper airway patency during loss of consciousness and sleep,²⁷¹ and may be a viable adjunct to achieve postoperative pain therapy in patients at high risk of airway collapse.

Nondepolarizing neuromuscular blocking drugs should be titrated optimally by quantitative neuromuscular transmission monitoring to avoid residual neuromuscular blockade, which increases the risk of postoperative respiratory complications.²⁷²⁻²⁷⁵ The effects of neuromuscular blockade should be reversed only when present, because inappropriate reversal can impair upper airway function in animals and humans.²⁷⁴⁻²⁷⁶

BOX 10.2 Special Sleep-Disordered Breathing Anesthesia Bundle: Special Procedures Performed During Anesthesia in Patients With Diagnosed Sleep-Disordered Breathing and Positive Airway Pressure or Noninvasive Ventilation Treatment

Preanesthesia Period

- Consider regional anesthetic techniques that minimize the chance of postoperative sedation.

Induction Strategy

- Monitoring: capnogram, tidal volume measurement
- Sniffing position
- Reverse Trendelenburg position
- Consider intubation without nondepolarizing NMBA; consider succinylcholine.
- Triple airway maneuver with two hands
- Utilize lung recruitment maneuvers immediately after intubation and apply PEEP for maintaining lung volume during surgery.
- PCV with PEEP
- Short-acting anesthetics and narcotics preferred
- Avoid high-dose steroidal NMBA.
- Use neuromuscular transmission monitoring.

Intraoperative Management

- Whenever possible, use of sedatives and narcotics should be reduced.
- Agents with reduced impairing effect on upper airway patency might be considered (e.g., ketamine, pentobarbital).

- Neuromuscular blockade should be monitored.
- Residual neuromuscular blockade should be reversed.

Extubation and Postanesthetic Care Unit

- Patient should be able to cooperate before extubation. Consider positioning of patients in PACU bed: upper body should be elevated by 45 degrees; lateral position preferred to minimize gravitational effects on the upper airway.
- In case of impaired respiratory function, a plan needs to be defined and documented for monitoring and treatment, including the consideration of noninvasive ventilation.
- Patients will be discharged to an unmonitored environment or home when they meet discharge criteria:
 - Vital signs within 20% from baseline
 - Adequate treatment of nausea
 - Pain score $\leq 40\%$
 - Aldrete-score ≥ 8
 - Passed room air challenge test

Pain Therapy

- Consider nonsteroidal antiinflammatory drugs to reduce opioid use whenever possible, if not contraindicated.
- Use caution when combining opioids with sedatives or hypnotics.

NMBA, Neuromuscular blocking agents; PACU, postanesthesia care unit; PCV, patient-controlled ventilation; PEEP, positive end-expiratory pressure.

Postoperative Care. OSA occurs during sleep and sedation;^{277,278} therefore, tracheal extubation should be delayed until complete recovery of consciousness. Patients should be positioned with the upper body elevated by 45 degrees, which improves airway patency^{261,262} and functional residual capacity. The lateral body position is an alternative for patients who do not tolerate an elevated upper body position. NIV is a viable treatment of postoperative respiratory failure and helps prevent deoxygenation and the development of postoperative negative pressure pulmonary edema; it can also prevent the respiratory depressant effects of opioids.^{276,279}

Before discharge from the post anesthesia care unit, a room air-challenge test should be conducted. Several noninvasive monitoring devices have been developed for patients with and without risk for OSA. However, it is currently unclear if these devices can help prevent postoperative complications, and this requires further investigation.^{280,281}

Pain Therapy. Pain treatment is of special concern during the postoperative period. Opioids dose-dependently reduce respiratory drive, and special caution is recommended when opioids are combined with sedatives or hypnotics. If not contraindicated, regional anesthesia and nonopioid analgesics should be considered to reduce opioid use whenever possible. If opioids are used, CPAP treatment throughout the early postoperative period may be beneficial in patients with high risk for OSA.²⁸² Pain intensity typically decreases over time following surgery. Decreasing opioid doses are typically required to maintain the balance of decreasing and increasing effects on respiratory drive.

Patients with SDB being treated with CPAP at home should proceed with these treatments throughout the perioperative period. Sufficient functioning of the treatment device needs to be checked preoperatively (see Box 10.2).

KNOWLEDGE GAPS AND FUTURE RESEARCH

Although a bold body of evidence on the effect of sleep-disordered breathing throughout the perioperative period has been published during the last decades, there are still large gaps of knowledge.^{283,284} Respiratory, cardiovascular, and neuropsychological complications, as well as unfavorable clinical outcomes, may be more or less common in patients with OSA. However, there are many comorbidities that are potential confounders, such as obesity, diabetes, dyslipidemia, coronary artery disease, and increasing age. OSA is a heterogeneous disease, and detailed physiologic phenotyping could identify pathophysiologic mechanisms that increase or decrease risk of perioperative complications of OSA. Multiple mechanistic pathways that can lead to OSA (e.g., compromised anatomy, dilator muscle dysfunction, low arousal threshold, elevated loop gain, inadequate lung volume tethering, and vascular leak) are potentially relevant in a perioperative setting.¹²⁵ In-depth physiological studies are required until phenotyping can be accomplished with less time-intensive studies or through PSG. Noninvasive and readily available physiological studies or clinical prediction models that can characterize patients who have high-risk phenotypes of OSA are an important area of future research.

In this context, adequate screening tools for OSA during the perioperative period need to be identified. As stated

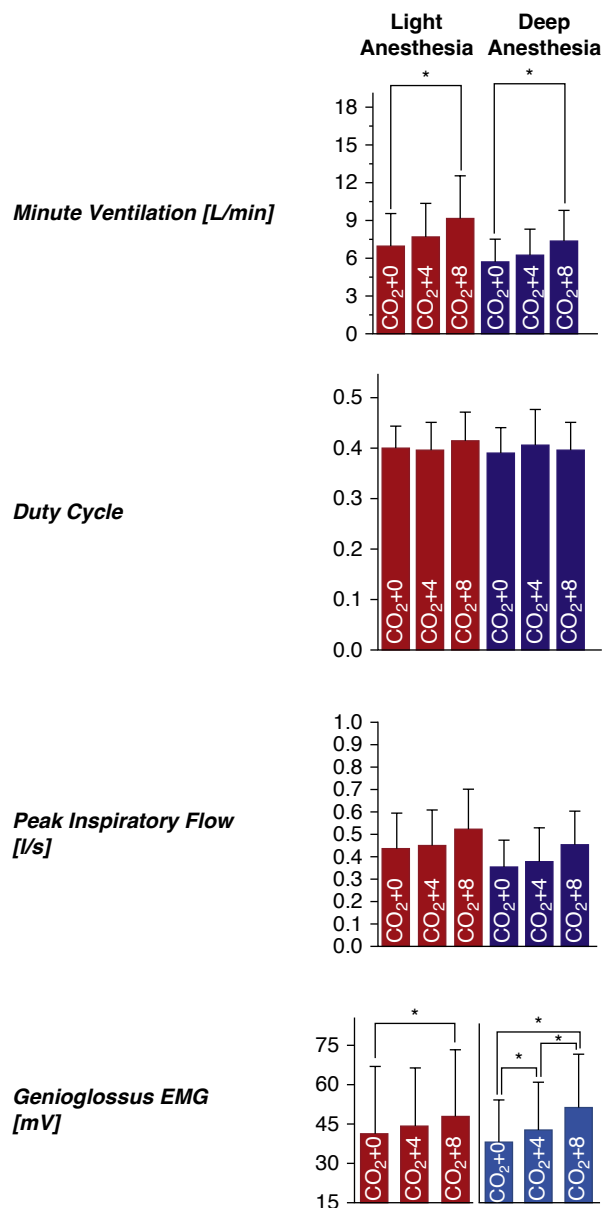


Fig. 10.14 Effect of carbon dioxide (CO₂) level on genioglossus activity and respiratory function during anesthesia. Insufflation of CO₂ to an increased pCO₂ (+4 or +8 mm Hg) increases minute ventilation, peak inspiratory flow and upper airway dilating muscle activity (e.g. genioglossus muscle) in a dose-dependent fashion during deep (blue bars) and light (red bars) anesthesia. (Modified from Ruscic KJ, Bøgh Stokholm J, Patlak J, et al. Supplemental carbon dioxide stabilizes the upper airway in volunteers anesthetized with propofol. *Anesthesiology*. 2018;129(1):37–46.)

previously, PSG is often challenging to schedule before surgery, and ambulatory technologies may be more useful and should be studied. It might even be more important to identify patients at increased risk of postoperative complications due to specific type of surgery, required mode and depth of anesthesia and opioids, as well as other comorbidities, rather than the presence of SDB per se. It is currently unclear to which extent the risk for perioperative complications associated with SDB are due to or modulated by these coexisting factors. In addition, more severe OSA may be developed by some patients with no or mild OSA prior to anesthesia. Challenging factors are opioids, body position, or intravenous fluids.

Some studies found early postoperative monitoring to detect patients with high risk for such aggravation of OSA severity. Additional research is needed to determine if and how postoperative monitoring is able to help stratify the risk for postoperative OSA. During the last few years, different approaches were made to improve postoperative management of patients with OSA (as discussed previously). However, open questions remain: What is the role of supplemental oxygen postoperatively? What are the barriers to CPAP adherence in the perioperative setting? Can patient adherence and positive airway pressure effectiveness be increased with educational resources? Should other respiratory support interventions be examined either in isolation or bundled with positive airway pressure? Is there a role for targeted positive airway pressure therapy, even with CPAP naïve patients in very high risk groups? What are the medicolegal liability and patient safety factors to consider when starting CPAP in the hospital?²⁸⁵

Future research should aim to answer these questions to further improve the perioperative management of patients with sleep disordered breathing.²⁸³ Testing a bundled approach to care (e.g., an algorithm of care including monitoring, positive airway pressure, education, and other respiratory supports) rather than each component individually might be more useful in initial clinical trials, as this may be more likely to be effective in improving outcomes.

The incidence of perioperative complications is low, and collaborative research networks are needed to identify a sufficient number of patients with OSA and answer the important research questions on perioperative outcomes. Patients with OHS might be at high risk. A minimal set of data elements required in prospective cohort studies should be identified to facilitate multisite collaboration and metaanalysis of independent studies. Case-control studies might also be nested within larger cohort studies. This would facilitate the choice of an appropriate control group. These studies may improve risk stratification and lead to novel targeted therapies. In addition, the identification of interventions that can optimize safety in this population is also important. Initial intervention trials should focus on patients at highest risk for sleep disordered breathing.²⁸³

Sleep and Sedation in the Intensive Care Unit

Impaired sleep is common in patients in the ICU,²⁸⁶ with multiple contributing factors including noise, light, interruptions for medical procedures, and administration of fluids and nutrition, as well as endogenous factors such as pain, anxiety, and inflammation. Sleep duration in the ICU is variable depending on environmental conditions and the severity of the patient's disease.²⁸⁷ Sleep fragmentation occurs consistently during sedation and mechanical ventilation.^{288,289} The duration of deep NREM and REM sleep as well as circadian rhythms are severely reduced or abolished in ICU patients.²⁹⁰⁻²⁹² About 50% of sleep in these patients occurs during daytime hours.^{286,287} Altered EEG activity, not fitting into the standard criteria for sleep and wakefulness, was reported to occur in up to 70% of ICU patients who were receiving sedatives or being mechanically ventilated.²⁹³ As a result, alternative classifications for sleep in the ICU have been developed;²⁹⁴ however, these scoring classifications have not yet been validated.

TABLE 10.7 List of Studies Investigating Alteration of Circadian Rhythm and Sleep in Intensive Care Unit Patients

Measurement Method	Interactions	Reference
PSG studies	■ Increased arousal during nighttime and disrupted sleep, due to: ■ noise ■ dim light ■ medical intervention ■ Decrease N3- and REM-sleep ■ Increase arousals and awakenings ■ Multiple discrete sleep episodes per 24 h ■ Atypical EEG signal: ■ disrupted sleep ■ nonphysiologic sleep ■ architecture, or evidence of coma	292,296 302-304 287,290,292,294
Melatonin blood levels and urinary 6-sulfatoxymelatonin urine levels	■ Increasing chronodisruption with disease severity (rehabilitation patients, ICU patients, ICU patients with severe sepsis) ■ Delayed melatonin peak at night ■ Reduced physiologic melatonin secretion at night, due to: ■ sedation ■ mechanical ventilation ■ increased arousal ■ Increased melatonin levels during the day, due to: ■ dim light ■ severe sepsis	292,304-312 305,308,309 306,311
Blood pressure variation	■ Reduced nocturnal blood pressure dip	305
Body temperature variation	■ Reduced temperature variation ■ Altered circadian rhythm with nadir dispersed widely across 24 h	23,305,313
Plasma cortisol level	■ Abolished circadian rhythm ■ Delayed peak of cortisol levels	305,310

EEG, Electroencephalography; ICU, intensive care unit; N3, sleep stage 3; PSG, polysomnography; REM, rapid eye movement.

NOISE AND LIGHT EXPOSURE

ICUs are among the most noisy areas in hospitals, with sound levels often exceeding those recommended by the World Health Organization (<45 dB).^{289,295} Elbaz and associates²⁹⁶ reported an increased incidence of arousals at sound levels above >77 dB in stable, mechanically ventilated ICU patients during the weaning phase. In the ICU setting, alarms from medical and monitoring devices were the most significant source of noise.

In addition to noise, changes in light exposure can alter sleep patterns and circadian rhythms. ICU patients are often exposed to dim light throughout the 24-hour cycle, resulting in increased light levels during the night and insufficient light exposure during the daytime.^{297,298} This hinders the alignment (photoentrainment) of the internal circadian clock to the external 24-hour rhythm, potentially resulting in the disrupted sleep-wake pattern seen in many ICU patients.²⁹⁹

Impaired rhythmicity in melatonin secretion pattern in critically ill patients was reported in several studies (Table 10.7 for overview), and multiple factors influence melatonin rhythm in the ICU. Melatonin levels decrease with increasing age,³⁰⁰ making older ICU patients more sensitive to circadian rhythm disruptions. Opioids increase melatonin levels throughout the 24-hour period.³⁰¹

Similarly, β -agonists, including vasopressors, positive inotropes, and aerosolized albuterol, increase melatonin levels, whereas β -blockers decrease melatonin secretion. Also, endogenous production of catecholamines may induce periodic melatonin increases in patients with critical illness.³⁰⁵ Benzodiazepines have inconsistent effects on melatonin.^{314,315}

These interactive effects of critical illness and ICU treatment should be acknowledged while caring for ICU patients. Reduction of light and noise exposure during the night, and increased light levels during the day are recommended to improve sleep. In addition, earplugs and eye masks have been shown consistently to improve sleep quality in ICU patients and possibly reduce the risk of complications.³¹⁶⁻³¹⁹

MEDICATION AND MEDICAL PROCEDURES

Interruptions of sleep due to medical or nursing procedures are environmental factors in the ICU. Studies in different ICU settings found high numbers of interventions to be performed during the nighttime, ranging up to 60 interventions per night.³²⁰⁻³²²

Both NIV and ventilation via breathing tube impair sleep architecture and quality.^{288,323} It is unclear how the issue of ICU-associated circadian disruption can be prevented or treated. Data suggest that weaning procedures during the daytime combined with assist control ventilation during the nighttime improve sleep, expressed as reduced numbers of arousal in patients requiring long-term ventilation.²⁹¹

SLEEP AND SEDATION IN THE INTENSIVE CARE UNIT

Sedation is frequently required in the ICU to treat excessive agitation or to fulfill specific therapeutic needs; however, heavy sedation is associated with prolonged ICU and hospital stays.³²⁴ GABA receptor agonists, such as propofol and benzodiazepines, may reduce REM sleep.^{325,326}

Pain is a common experience for most patients in the ICU,³²⁷ and failure to recognize pain can result in excessive administration of sedatives.³²⁴ However, opioids have been shown to decrease REM-sleep and worsen sleep disorders, such as OSA in ICU patients.^{131,328} Accordingly, an aggressive approach to managing pain while minimizing opioid utilization has been recommended.³²⁹

PHARMACOLOGICAL TREATMENT OF SLEEP DISTURBANCES IN INTENSIVE CARE UNIT PATIENTS

Medications currently used for the treatment of insomnia in the ICU interact with the GABA system. GABAergic neurotransmission has multiple functions throughout the brain, leading to daytime sedation, confusion, anterograde amnesia, and delirium.³³⁰⁻³³²

Melatonin has been selected as alternative sleep-promoting medication for critically ill patients in some countries. However, studies on the use of melatonin for the treatment of sleep disturbances and potential complications (e.g., delirium) in ICU patients have produced conflicting results. Administration of melatonin improved subjective and objective sleep quality in a small number of studies.³³³⁻³³⁵ Melatonin may have side-effects such as dizziness, nausea, drowsiness, hypotension, and headache, potentially critical in the ICU setting. Recently melatonin receptor agonists have been developed and marketed for patients with circadian rhythm disorders.^{336,337} These drugs may have fewer off-target effects than melatonin, but studies on their application in the ICU are very limited. It is therefore difficult to recommend treatment with melatonin agonists of any kind for ICU patients.³³⁸

The orexin receptor antagonist suvorexant has been approved by the FDA for the treatment of primary insomnia and might be another promising new approach for the treatment of insomnia in ICU patients. In a 4-week study in 254 patients with primary insomnia, suvorexant significantly improved sleep efficiency from the first night until the end of the study compared with placebo in a dose-dependent manner. In addition, suvorexant significantly improved wake after sleep onset and reduced sleep latency,³³⁹ without an apparent delirium-promoting effect.³⁴⁰ Additional research in the efficacy and safety of orexin receptor antagonists with critically ill patients is needed.

CONSEQUENCES OF SLEEP DISTURBANCE IN ICU PATIENTS

Compelling evidence links good sleep to proper metabolic, endocrine, immune, and neurobehavioral function.³⁴¹⁻³⁴⁵

Regardless of controversies, the importance of a sufficient amount of good quality sleep for proper immunologic function, as well as the increased susceptibility for infection due to sleep loss, has gained increased acceptance.³⁴⁶ Sleep deprivation during influenza vaccination delays the increase of antibody titers.^{347,348} Sleep deprivation and insomnia is associated with decreased phagocytic activity of lymphocytes and natural killer T-cells, as well as lower levels of interleukin 2³⁴⁹ and interleukin 7.³⁵⁰ Interleukin 7 facilitates the transition of CD8p

effector to memory T-cells and lengthens survival of the T-cell memory cells.³⁵¹ Prolonged sleep deprivation, as may occur in long term ICU patients, may induce a persistent production of proinflammatory cytokines, producing a chronic low-grade inflammation, as well as immunodeficiency.³⁴⁸

Even short sleep deprivation in otherwise healthy young subjects has been shown to be associated with impaired glucose tolerance and insulin resistance, similar to levels usually seen in patients in the early course of diabetes mellitus.³⁵² In addition, sleep deprivation can induce the onset of a catabolic state: increased oxygen consumption, carbon dioxide production, and catecholamine levels, likely as signs of a stress response.³⁵³ Changes in the activity of the hypothalamo-pituitary-adrenal axis with altered plasma levels of cortisol and thyrotropin,³⁵⁴ as well as increased levels of inflammatory cytokines, are seen during sleep deprivation.³⁵⁵

It is likely that postoperative sleep deprivation also affects respiratory function, characterized as increased vulnerability to respiratory muscle fatigue,³⁵⁶ decrease in ventilatory response to hypercapnia,³⁵⁷ and greater upper airway collapsibility.³⁵⁸

Cognitive impairment after even short-term sleep deprived conditions is intuitive. Sleep deprivation causes bidirectional changes in brain activity and connectivity, thereby mainly affecting attention and working memory, and an increased vulnerability to delirium in critically ill patients.³⁵⁹⁻³⁶¹

In addition, recent studies found beneficial effects of realignment of the sleep-wake cycles by pharmacologic and chronotherapeutic methods on the risk for delirium in these patients.³⁶²⁻³⁶⁴ However, sleep-promoting agents per se can increase the vulnerability to delirium, and the optimal sleep-promoting agent that also minimizes the risk of delirium needs to be identified.

PERIOPERATIVE MANAGEMENT OF OTHER SLEEP DISORDERS

Narcolepsy

Narcolepsy is a neurologic sleep disorder with a prevalence of 0.05% to 0.8% in most ethnic groups.^{365,366} It is characterized by excessive daytime sleepiness, involuntary daytime sleep episodes, disturbed nocturnal sleep, and sleep-associated muscular hypotension. Narcolepsy is divided into narcolepsy with or without cataplexy (sudden loss of muscle tone without loss of consciousness).³⁶⁷

An autoimmune pathology against hypothalamic orexin neurons may be involved in the pathogenesis of narcolepsy.³⁶⁶ A mutation of the HCRT receptor 2 or loss of HCRT neurons have been shown to cause a narcolepsy-like state in animals,³⁶⁸ and the deficiency of HCRT has been shown to be associated with narcolepsy in humans.³⁴ HCRT is involved in control of several biological functions such as feeding, cardiovascular regulation, upper airway stability, pain, locomotion, stress, and addiction.^{369,370} Environmental factors play a key role in the pathogenesis of the disorder, as the disease concordance rate between monozygotic twins is only 20% to 35%.

Treatment of narcolepsy consists of behavioral treatment, as well as pharmacologic treatment for (1) daytime

sleepiness and (2) cataplexies. Periodic and regular sleep times and scheduled daytime naps are recommended. Pharmaceutical treatments of daytime sleepiness include amphetamines, methylphenidate, modafinil, or selegiline (also effective for treatment of cataplexy), and cataplexies can be treated using tricyclic antidepressants, selective serotonin reuptake inhibitors, or γ -hydroxybutyrate/sodium oxybate. Medical treatment should always be accompanied by behavioral therapy.

Delayed emergence from anesthesia, postsurgical hypersomnia, and apneic episodes are in part related to an increased sensitivity to anesthetic drugs in narcolepsy patients.³⁷¹⁻³⁷³ Monitoring of anesthetic depth might be recommended in this patient population. Medical treatment of narcolepsy should be maintained during the preoperative period.^{374,375} The most commonly used treatment of daytime sleepiness is modafinil, which acts via dopaminergic pathways and accelerates emergence from anesthesia.^{376,377} Avoidance of sedative premedication and consideration of regional anesthesia may be indicated.³⁷⁸ Of note, cataplectic events can also occur during regional anesthesia.³⁷⁹

Restless Legs Syndrome and Periodic Limb Movement Disorder

Restless legs syndrome, or Ekbom syndrome, is a neurologic disorder with a prevalence of 2% to 5% as defined by four cardinal features, including (1) the urge to move the limbs, usually associated with paresthesias or dysesthesias, (2) aggravating effects of rest, (3) ameliorating effects of physical activity, and (4) symptoms that worsen during the course of day, with a peak during the evening or at night. Patients with restless legs syndrome usually also complain about sensory symptoms in the legs.

Isolated periodic limb movement during sleep is a rare symptom, commonly referred to as *periodic limb movement disorder*. The characteristic periodic episodes of repetitive limb movements during sleep occur most often in the lower extremities, but occasionally in the upper extremities. These movements can be associated with frequent arousals leading to sleep disruption, causing excessive daytime sleepiness, which often is the only symptom reported by the patients themselves in most cases.³⁸⁰

Symptomatic restless legs syndrome can occur in patients with iron deficiency and uremia, during pregnancy,³⁸¹ or during the use of neurotropic medications (dopamine antagonists, neuroleptics, selective serotonin reuptake inhibitors, tricyclic antidepressants, antihistamines, caffeine, alcohol, nicotine). Although daytime symptoms of restless legs syndrome can be sufficiently diagnosed clinically (clinical examination in combination with standardized questionnaires), a PSG is recommended to rule out SDB, especially in patients complaining of daytime sleepiness or sleep fragmentation.

According to the most recent guidelines of the AASM, the first-line treatment of restless legs syndrome should consist of dopamine agonists (i.e., ropinirole and pramipexole) in the evening. Furthermore, gabapentin enacarbil, levodopa with dopa decarboxylase inhibitor, or opioids can be used. Medication impairing the dopaminergic system (dopamine antagonists, neuroleptics, selective serotonin reuptake

inhibitors, and tricyclic antidepressants, antihistamines, caffeine, alcohol, and nicotine) should be avoided, if possible.

Exacerbation of restless legs syndrome may occur after general anesthesia,³⁸² and the urge to move the limbs may be misinterpreted as agitation or delirium.³⁸³ The first manifestation of restless legs syndrome may occur after spinal anesthesia³⁸⁴ or general anesthesia,³⁸⁴ and the prevalence of restless legs syndrome after surgery seems to be higher than expected in this population.³⁸² To prevent perioperative exacerbation of symptoms, patients with restless legs syndrome should be scheduled for surgery early in the day. Restless legs syndrome medication should be continued until the day of surgery, whenever appropriate. Drugs that block the central dopamine transmission, such as neuroleptics, should be avoided. In contrast, ketamine might be the superior anesthetic drug for patients with restless legs syndrome.³⁸⁵ In addition, intravenous or subcutaneous opioids and benzodiazepines during and after the surgery procedure might be beneficial in patients with restless legs syndrome. The best way to provide symptom relief in patients with restless legs syndrome might be early mobilization after surgery. In patients who are not eligible for mobilization, compression treatment³⁸⁶ or intravenous administration of magnesium³⁸⁷ and physostigmine³⁸⁸ have been shown to lead to relief of restless legs syndrome symptoms. Iron and ferritin blood levels should be monitored closely before, during, and after surgery, especially in surgical cases with iron loss (i.e., bleeding), to prevent symptomatic restless legs syndrome symptoms.

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